

primat

A Python package (with a compiled C engine) for precision Big Bang Nucleosynthesis

User guide and physics reference

Documentation for primat version 0.3.3

Cyril Pitrou

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Abstract

primat (formerly PyPRIMAT) is a re-implementation of the PRIMAT code for precise Big Bang Nucleosynthesis (BBN), distributed as a single pip-installable package with two interchangeable backends: a fast compiled C engine (the default) and a pure-Python implementation (used as a fallback and for development/extensibility, e.g. custom reaction networks via the GUI). Either backend integrates the coupled equations governing (i) the cosmological background – photon and neutrino temperatures and the scale factor – and (ii) a nuclear reaction network, to predict the primordial abundances of ^1H , D , ^3He , ^4He , ^7Li and heavier nuclides. This document is meant to be *self-contained*: the main text shows how to *use* the code (Sec. 1) and gives the working equations for the three physical ingredients – the plasma thermodynamics (Sec. 2), the $n \leftrightarrow p$ weak interactions (Sec. 3) and the thermonuclear network (Sec. 4) – together with the run-time options that control them, followed by a section on the sensitivity of the abundances to the inputs (Sec. 5). All the lengthy formulas needed to actually reproduce the code – the full QED plasma corrections, every correction to the weak rates beyond the Born approximation, and the detailed-balance/NSE derivation – are collected in the appendices, so that the algorithms can be rewritten from this document alone. Equations follow the review of Pitrou, Coc, Uzan & Vangioni, *Physics Reports* **754** (2018) 1 [1] (cited as “Phys. Rep.”), and references therein, which the reader should consult for derivations. Figures are produced by the companion notebook `primat_doc_figures.ipynb`. This document describes **primat version 0.3.3**: the package was renamed from `pyprimat` to `primat` and gained the compiled C backend, a Streamlit GUI with a custom-network builder, Monte-Carlo rate-uncertainty propagation, and a data-directory overlay mechanism since the v0.1.0 edition of this document; see Sec. 1 for all of these.

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1 Quick overview: using the code

`primat` can be driven in four equivalent ways – a command-line tool, a Python API, a graphical interface, or (behind all three) the low-level backend dispatcher – all of which build the same parameter dictionary and ultimately run the same physics. Run from the repository root so that the data files under `primat/data/` resolve, or use a `pip` installed copy (data ships inside the package, see the “Rates directory resolution” paragraph below). Every code example below uses the shipped *compiled C engine*, the default and fastest of the two interchangeable backends described in §1.2.

1.1 Command-line interface

The `primat` console script exposes the most commonly varied options:

```
primat --Omegabh2 0.022425 --network large --amax 8
```

which prints a short summary:

```
-----
PRIMAT results at T = 0.001 MeV
-----
Neff = 3.04397730
YP (BBN) = 0.24700459
```

```

YP (CMB) = 0.24567826
He4/H = 8.2013888e-02
D/H = 2.4356594e-05
He3/H = 1.0396011e-05
He3/He4 = 1.2675916e-04
Li7/H = 5.503013e-10
Li6/Li7 = 1.418195e-05
--- running time: 0.69 seconds ---

```

The main named flags are `--Omegabh2`, `--DeltaNeff`, `--network` (small/small_parthenope/large or any custom network file under `data/nuclear/networks/`), `--amax` (restrict any network to nuclides of mass number $\leq A$; e.g. the example above reproduces the old “medium” network’s 68 reactions, §4.4), `--numerical_precision`, `--backend` {auto,c,python} (§1.2), `--mc N` (also run an N -sample Monte-Carlo rate/ τ_n -uncertainty propagation and print each observable as “value +/- sigma”, §1.2), `--json` (dump the full result dictionary instead of the summary), `--verbose`, `--cache-info/` `--cache-clear` (inspect/clear the cached weak-rate files), and `--credits/--version`. Any other PRIMATConfig key, including a per-reaction `p_<reaction>/delta_<reaction>` rate variation (§4), can be set with the generic escape hatch `--set KEY=VALUE` (repeatable), e.g. `--set T_end_MeV=1e-4 --set decay_era=True`.

1.2 Python API

For full control, build a parameter dictionary and call PRIMAT directly; every key of `config.DEFAULT_PARAMS` (95 keys) is accepted.

```

from primat import PRIMAT

result = PRIMAT({"Omegabh2": 0.022425, "network": "large", "amax": 8}).solve()
print(f"YP (BBN) = {result['YPBBN']:.8f}") # 0.24700444
print(f"D/H = {result['DoH']:.7e}") # 2.4356398e-05
print(f"Neff = {result['Neff']:.8f}") # 3.04397730

```

`solve()` (equivalently `primat_results()`, kept for backwards compatibility) returns a dictionary with `Neff`, the helium mass fractions `YPBBN/YPcmb`, the ratios `He4oH`, `DoH`, `He3oH`, `He3oHe4`, `Li7oH` (and, for networks tracking them, `Li6oLi7` and CNO yields), and the neutrino observables `Omeganurel/OneOverOmeganunr`. PRIMAT is a thin facade over the background and nuclear-network classes: with the *Python* backend (`force_backend="python"`, or automatically whenever a Python-only feature such as `extra_rho/background=/decay_era` is used) the instance also exposes the background history and abundance trajectories as attributes used throughout the companion notebook: the temperature vectors `_Tg_vec`, `_Tnue_vec`, the weak-rate interpolants `_nT0p_frwr`, `_nT0p_bkwr` (functions of T_γ in *kelvin*), the compiled forward reaction rates `nucl.<reaction>_frwr` (function of T in kelvin, internally $T_9 = T/10^9$), the abundance interpolant `_Y_of_t`, and the final values `_Y_final`. The C backend does not expose these intermediate attributes (only the final result dict and, optionally, the time-evolution arrays below), which is why the figures in this document are generated with the Python backend even though it is no longer the default for routine runs.

The `primat.backend` dispatcher. `PRIMAT(...)/.solve()` always uses the pure-Python engine; day-to-day scripts should instead call the dispatcher directly,

```

from primat.backend import run_bbn
result = run_bbn(params={"Omegabh2": 0.022425}, force_backend="auto")

```

which picks the compiled C extension (`primat._primat_c`, wrapping the standalone `primat-c/C99` port) when it built successfully (`primat.backend.HAS_C_BACKEND`), falls back silently to Python otherwise, and returns the identical result-dict shape either way (the two backends are

held to a relative 5×10^{-5} budget on D/H for the same inputs, against a measured gap of 6.6×10^{-6} ; that residual is each backend’s own ODE-tolerance error, and N_{eff} agrees to every digit — see `tests/test_backend_parity.py`). Pass `force_backend="c"` to require the compiled engine (raises if it is unavailable or if a Python-only feature is requested) or `force_backend="python"` to force the reference implementation. Custom reaction networks (§1.2), the full time-evolution schema (below), `extra_rho` and `decay_era` are supported identically on both backends, and incremental MC sample reuse (`prev=`, §1.2) works within either backend — but not across the two, whose random streams differ. Only `background=` (a custom `Background` object, which cannot cross the C ABI) is Python-only.

Monte-Carlo rate-uncertainty propagation. Every tabulated reaction rate and τ_n carries a 1σ uncertainty (§4); `primat.backend.run_mc` (or the CLI’s `--mc`) draws N independent samples — each reaction’s $p_{\langle \text{reaction} \rangle} \sim \mathcal{N}(0, 1)$ and $\tau_n \sim \mathcal{N}(\tau_n, \sigma_{\tau_n})$ — re-solves BBN for each, and returns an `MCResult` with the central (unperturbed) value, sample mean/std, and every individual sample for each requested observable:

```
from primat.backend import run_mc
mc = run_mc(200, params={"network": "large", "amax": 8}, seed=0)
print(f"D/H = {mc['DoH'].central:.7e} +/- {mc['DoH'].std:.2e}")
```

The C backend runs one pthread per sample with an independent xoshiro256** stream (not bit-for-bit comparable to the Python/NumPy path, only statistically); both cap each reaction’s sampled rescaling factor to `mc_rate_rescale_cap` (default 30, §4) to keep a handful of poorly-constrained placeholder-uncertainty rates from dominating the variance with unphysical outliers.

Custom reaction networks. Both backends accept a `custom_network` override (remove/replace/add reactions and swap in an alternate rate table for any reaction), built either programmatically (`primat.network_data.UpdateNuclearRates`) or through the GUI’s “Import/Create custom network” popups below, which can export the result as a re-importable `.zip`.

Output files and the unified time-evolution schema. `output_time_evolution=True` populates (in memory, on both backends) an `EvolutionResult` with the unified schema documented in `primat/evolution.py`’s module docstring — `t_s`, `a`, `T_gamma_MeV`, the three flavour neutrino temperatures, and every tracked nuclide’s `Y_{\langle \text{nuclide} \rangle}` column, one row per output time step — and additionally writes it to `output_file` (default `results/output_tables.tsv`) if that is set (`output_file=None` skips the disk write, e.g. for the GUI, while still populating the in-memory result). `output_final_result=True` writes the final mass fractions to `output_final_file`; `output_background_evolution/` `output_mc_samples` similarly control the background-only and MC-samples TSV dumps.

Rates directory resolution. Two `PRIMATConfig` fields let a run substitute alternate data without touching the installed package: `data_dir` (a full replacement of the shipped `primat/data/` tree — must contain `NEVO/`, `nuclear/`, `csv/` and `cache_plasma_weak/{weak,plasma}/`) and `user_nuclear_dir` (an *additive* overlay for nuclear networks and rate tables only, checked before the shipped tree so built-in networks like `small` remain available). Both are validated as existing directories at `PRIMATConfig` construction time and are ordinary string parameters on the C side too (`--data_dir/CPRIMAT_DATA_DIR` for the standalone C CLI).

1.3 Graphical interface

After `pip install ".[gui]"`, a browser-based Streamlit app is available:

```

primat-gui # console-script launcher
primat-gui --backend python # force the Python backend (testing)
streamlit run primat/gui/app.py # from a source checkout

```

It exposes the parameters as form fields, shows the final abundances and an interactive abundance-evolution plot, and additionally offers, in the “Nuclear reactions” sidebar group, a single **Manage networks** button – the one gateway to listing, selecting, removing or renaming a network built this session, loading one from a .zip, or opening **Create new network**, which lets a user start from any named network, toggle reactions in/out grouped by mass-number category, substitute or upload an alternate rate table per reaction, override a decay rate and add brand-new reactions, then save the result as a re-importable .zip or run BBN with it (§1.2); a results tab exposes the same Monte-Carlo uncertainty propagation as `--mc` (§1.2).

1.4 Reading the numerical output

Several physical corrections shift N_{eff} at the 10^{-2} – 10^{-3} level, so abundances should be reported with enough significant figures: at least 8 decimals for N_{eff} and Y_P , 7 for D/H, and 6 for ${}^7\text{Li}/\text{H}$. A standard small-network run with $\Omega_b h^2 = 0.022425$ gives $Y_P \simeq 0.24700$, $\text{D}/\text{H} \simeq 2.4350 \times 10^{-5}$ and $N_{\text{eff}} \simeq 3.04398$.

2 Plasma thermodynamics

2.1 Background equations

The BBN background is a hot plasma of photons, electrons/positrons and three neutrino species in a spatially-flat Friedmann–Lemaître spacetime. Each species is described by a distribution function $f(t, p)$ obeying the Boltzmann equation in an expanding universe (Phys. Rep. Eq. 12),

$$\frac{\partial f}{\partial t} - H p \frac{\partial f}{\partial p} = C[f], \quad (1)$$

whose velocity moments give the conservation laws for number density and energy density (Phys. Rep. Eqs. 13, 16),

$$\dot{n} + 3Hn = \mathcal{J}, \quad \dot{\rho} + 3H(\rho + P) = \dot{q}, \quad (2)$$

with $\mathcal{J}$ and $\dot{q}$ the collisional source terms. The thermodynamic quantities are momentum integrals of f ; for a species of degeneracy g and mass m , writing $x \equiv m/T$, they reduce to the Fermi–Dirac (+) / Bose–Einstein (−) integrals $I_{\pm}^{(n,m)}$ defined in App. A (Phys. Rep. App. A):

$$\rho = \frac{g T^4}{2\pi^2} I_{\pm}^{(2,1)}(x), \quad P = \frac{g T^4}{6\pi^2} I_{\pm}^{(0,3)}(x), \quad s = \frac{\rho + P}{T}. \quad (3)$$

The photon gas ($g = 2$, $m = 0$) has $\rho_{\gamma} = (\pi^2/15) T_{\gamma}^4$. The expansion rate is fixed by the Friedmann equation (Phys. Rep. Eq. 9),

$$H^2 = \frac{8\pi G}{3} \rho_{\text{tot}}, \quad \rho_{\text{tot}} = \rho_{\gamma} + \rho_{e\pm} + \sum_{\nu} \rho_{\nu} + \delta\rho_{\text{QED}} + \dots, \quad (4)$$

and the scale factor $a(T_{\gamma})$ follows from conservation of the comoving electromagnetic-plasma entropy $s a^3$ (Phys. Rep. Eqs. 21–24).

2.2 Solving the background: the two ODEs $a(T_{\gamma})$ and $t(a)$

The code reduces the background to two ordinary differential equations, solved once at start-up (`PRIMAT._setup_background_and_cosmo`).

Scale factor from entropy conservation. Define the reduced plasma entropy $\mathcal{S}(T) \equiv \bar{s}_{\text{pl}}/\bar{s}_\gamma$ (with $\bar{s}_\gamma = 2\pi^2/45$, so $s_{\text{pl}} = \bar{s}_\gamma T^3 \mathcal{S}$); from Eq. (3), $\mathcal{S} = 1 + \frac{2}{\pi^2 \bar{s}_\gamma} [\frac{1}{3} I_+^{(0,3)}(x) + I_+^{(2,1)}(x)]$, running from $11/4$ at $T \gg m_e$ to 1 at $T \ll m_e$ (Phys. Rep. Eq. 28). Exact entropy conservation gives the algebraic relation $a(T) = a_0 T_0 / (T \mathcal{S}^{1/3})$ (Phys. Rep. Eq. 29). With incomplete neutrino decoupling the plasma loses entropy to the neutrinos at a dimensionless heating rate $\mathcal{N}(T)$ defined by $\dot{q}_{\text{pl}}/(HT) = -T^3 \mathcal{N}$ (Phys. Rep. Eq. 57), and the relation becomes the ODE actually integrated (Phys. Rep. Eq. 58),

$$\frac{d \ln a}{d \ln T} = - \frac{3 \bar{s}_{\text{pl}} + T d\bar{s}_{\text{pl}}/dT}{\mathcal{N} + 3 \bar{s}_{\text{pl}}}, \quad \bar{s}_{\text{pl}} \equiv \frac{s_{\text{pl}}(T)}{T^3}, \quad (5)$$

which reduces to $a \propto (T \mathcal{S}^{1/3})^{-1}$ when $\mathcal{N} = 0$ (the QED-corrected $\mathcal{S}^{\text{QED}}$ of App. B is used when QED is on). In the code $\bar{s}_{\text{pl}}$ and $d\bar{s}_{\text{pl}}/dT$ are returned together by `plasma.spl_and_dspl_dT`; Eq. (5) is `_dlnadlnT_NEVO`, integrated from T_{end} to T_{start} with `solve_ivp(method='LSODA')`, and inverted to $T(a)$. When `external_scale_factor=True`, $a(T)$ is instead read directly from the NEVO table's column $x \propto a$ (requires `incomplete_decoupling=True`; the two modes agree to $\sim 10^{-5}$ in $a(T)/t(T)$).

Time from the Friedmann equation. With $a(T)$ – hence $\rho_{\text{tot}}(T(a))$ – known, the Friedmann equation (4) is integrated as

$$\frac{dt}{d \ln a} = \frac{1}{H(a)}, \quad t_{\text{ini}} = \frac{1}{2H_{\text{ini}}} \quad (6)$$

(radiation-domination start), giving $t(a)$ and its inverse by interpolation (`_dtdlna`). All subsequent eras use the resulting $T(t)$, $a(t)$, $t(T)$. The plasma energy density entering H is $\rho_{\text{pl}} = \mathcal{E}(T) \bar{\rho}_\gamma T^4$ with $\mathcal{E} = 1 + \frac{30}{\pi^4} I_+^{(2,1)}(x)$ (plus the QED term $\delta g_\rho/2$), and the three neutrino flavours contribute $\rho_\nu = \bar{\rho}_\nu T_\nu^4$ each (Phys. Rep. Eqs. 30–31).

Baryon density and η_b . Baryons are cold, $\rho_b \propto a^{-3}$, so $n_b(a) = n_{b,0}/a^3$ with $n_{b,0} = \eta_b n_{\gamma,0}$ and the baryon-to-photon ratio $\eta_b = (\Omega_b h^2) \times C_\eta$, where the conversion factor $C_\eta = \rho_{100}^{\text{crit}}/(n_{\gamma,0} m_b)$ (`cfg.Omegabh2_to_eta0b`); numerically $\eta_b \simeq 6.09 \times 10^{-10} (\Omega_b h^2 / 0.02225)$ (Phys. Rep. Eqs. 33–35). This $n_b(a)$ sets the two-body fluxes $\Gamma = n_b \langle \sigma v \rangle$ of Sec. 4.

Optional energy-density components. Any extra contribution to ρ_{tot} in Eq. (4) enters through the `extra_rho` hook of PRIMAT (a list of functions $\rho(T_\gamma)$ [MeV⁴]; Python backend only, §1.2). Two are built in: a constant extra radiation density from ΔN_{eff} decoupled species (`plasma.rho_nu_extra`, $\rho = \frac{7}{8} \frac{\pi^2}{15} T_\nu^4 \Delta N_{\text{eff}}$), and Early Dark Energy (`fEDE`, `zEDE`, `wnEDE`; `PRIMAT._setup_EDE`) – a fluid peaking at redshift `zEDE` with energy fraction `fEDE` and equation-of-state index `wnEDE`. Both modify only $H(T_\gamma)$, hence the time–temperature relation.

2.3 Neutrino decoupling and N_{eff}

Neutrinos decouple just before e^+e^- annihilation. In the idealized *instantaneous* limit, entropy conservation gives the asymptotic ratio $T_\nu/T_\gamma \rightarrow (4/11)^{1/3} \simeq 0.7138$, and the radiation density is parameterised by the effective number of neutrinos (Phys. Rep. Eq. 62),

$$\rho_{\text{rad}} = \rho_\gamma \left[1 + \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} N_{\text{eff}} \right]. \quad (7)$$

In reality decoupling is *not* instantaneous: the highest-energy neutrinos are still coupled during annihilation and absorb a little of the released entropy, raising N_{eff} above 3. `primat` reads the precomputed non-instantaneous history $\{T_\nu^{(e,\mu,\tau)}(T_\gamma)\}$ from a NEVO table rather than recomputing the momentum-dependent Boltzmann hierarchy. Together with the QED plasma corrections (below) this gives the Standard-Model value $N_{\text{eff}} \simeq 3.044$ (Fig. 1).

2.4 QED plasma corrections

Electromagnetic interactions between the $e^\pm$ and the photon bath shift the effective electron and photon masses and hence the plasma equation of state (Phys. Rep. §II.E). To leading orders in the coupling the interaction-pressure correction is $\delta P = \delta P_d + \delta P_s$, with a dominant $O(\alpha)$ term and a subdominant two-loop exchange term; the explicit integrals (Phys. Rep. Eqs. 47–49) are given in App. B. The energy-density correction follows from the thermodynamic identity $\rho = -P + T dP/dT$, and the high-temperature limit is the constant

$$\delta g_\rho \simeq \delta g_P \simeq -\frac{25 \alpha_{\text{FS}}}{4\pi} \simeq -0.0145 \quad (8)$$

(Phys. Rep. Eq. 52). These corrections enter in four places (App. B): the plasma entropy $\mathcal{S}$ that sets $a(T_\gamma)$, the neutrino-temperature normalisation z_0 , the plasma energy density in Eq. (4), and – through the electron mass shift, which is bookkept as a finite-temperature weak-rate correction (App. C) – the weak rates.

2.5 In the code, with figures

Photon and neutrino thermodynamics are pure functions of temperature (`plasma.rho_g`, `plasma.rho_nu`); the $e^\pm$ quantities (`rho_e`, `p_e`) require the Fermi–Dirac integrals of Eq. (3) and are tabulated in an electron-thermo cache. The non-instantaneous $T_\nu(T_\gamma)$ history is read by `neutrino_history.py`; the QED corrections are loaded from `data/cache_plasma_weak/plasma/QED_*.txt` or computed analytically by `qed_pressure.py`; the Friedmann rate is assembled in `PRIMAT._Hubble`. Figure 1 shows the per-flavour temperature ratios, Fig. 2 the effective relativistic degrees of freedom $g_{*\rho}, g_{*s}$ through annihilation, and Fig. 3 the QED pressure correction.

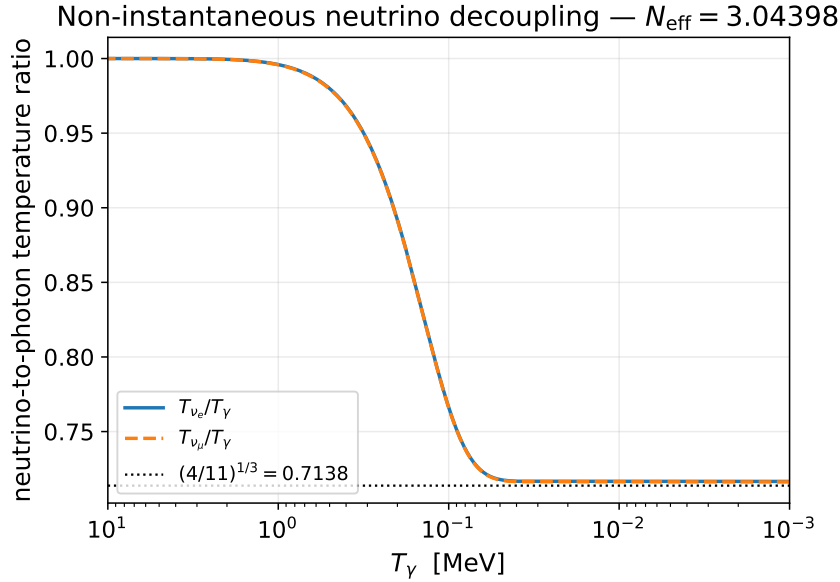


Figure 1: Per-flavour neutrino-to-photon temperature ratios from the NEVO non-instantaneous decoupling history. The dotted line is the instantaneous limit $(4/11)^{1/3}$; the residual offset is what raises N_{eff} above 3.

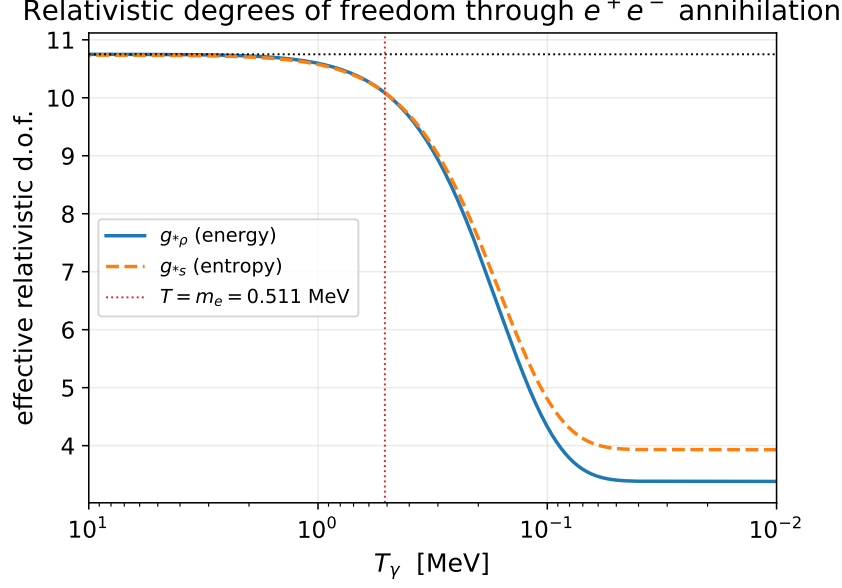


Figure 2: Effective relativistic degrees of freedom in energy ($g_{*\rho}$) and entropy (g_{*s}), dropping from 10.75 to $\simeq 3.36/3.93$ as e^+e^- annihilate around $T \sim m_e$, heating the photons relative to the neutrinos.

2.6 Associated options

Option	Default	Effect
<code>incomplete_decoupling</code>	True	Use the NEVO non-instantaneous $T_\nu(T_\gamma)$ table (else instantaneous).
<code>QED_corrections</code>	True	Include the QED plasma corrections (App. B).
<code>n_electron_table</code>	2000	Grid points for the $e^\pm$ thermodynamic tables.
<code>recompute_qed_corrections</code>	False	Recompute and overwrite <code>data/cache_plasma_weak/plasma/QED_*.txt</code> .
<code>external_scale_factor</code>	False	Read $a(T_\gamma)$ directly from the NEVO table instead of solving the ODE.
<code>DeltaNeff</code>	0	Extra decoupled radiation added to ρ_{tot} .
<code>munuOverTnu</code>	0	Neutrino degeneracy $\xi_\nu = \mu_\nu/T_\nu$.
<code>data_dir</code>	None	Full-takeover replacement for the shipped <code>primat/data/</code> tree.
<code>user_nuclear_dir</code>	None	Additive overlay for nuclear networks & rate tables only.

3 Weak interactions

3.1 The six processes and their rates

The neutron-to-proton ratio is controlled by six weak processes related by crossing symmetry (Phys. Rep. Eq. 67),

$$n + \nu_e \leftrightarrow p + e^-, \quad n \leftrightarrow p + e^- + \bar{\nu}_e, \quad n + e^+ \leftrightarrow p + \bar{\nu}_e, \quad (9)$$

grouped into a forward rate $\Gamma_{n \rightarrow p}$ (sum of the three) and a reverse $\Gamma_{p \rightarrow n}$. Each rate is a phase-space integral over the weak matrix element with the appropriate Fermi–Dirac occupations and Pauli-blocking factors (Phys. Rep. Eq. 70; App. C, Eq. (27)). The neutron-proton density evolution is

$$\dot{n}_n + 3Hn_n = -n_n\Gamma_{n \rightarrow p} + n_p\Gamma_{p \rightarrow n}. \quad (10)$$

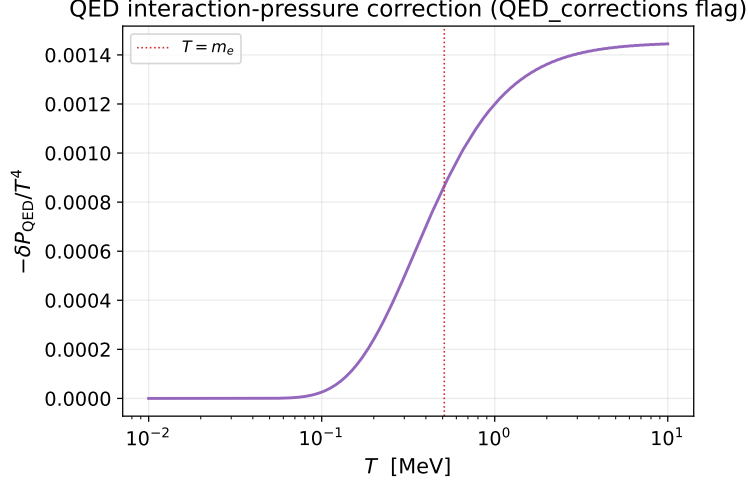


Figure 3: Leading QED interaction-pressure correction $-\delta P_{\text{QED}}/T^4$, peaking near $T \sim m_e$; its size is set by α_{FS}/π (toggled by `QED_corrections`).

3.2 Born approximation and calibration

In the infinite-nucleon-mass (Born) limit the matrix element collapses to a constant and the forward rate becomes the one-dimensional integral (Phys. Rep. Eqs. 77–83)

$$\bar{\Gamma}_{n \rightarrow p} = \mathcal{K} \int_0^\infty p^2 dp [\chi_+(E) + \chi_+(-E)], \quad \chi_\pm(E) \equiv (E_\nu^\mp)^2 g_\nu(E_\nu^\mp) g(-E), \quad (11)$$

with $E = \sqrt{p^2 + m_e^2}$, $E_\nu^\mp = E \mp \Delta$, $\Delta = m_n - m_p$, g and g_ν the electron and neutrino Fermi–Dirac distributions at T and T_ν , and the normalisation

$$\mathcal{K} = \frac{4 G_W^2 (1 + 3g_A^2)}{(2\pi)^3}, \quad G_W = G_F \cos \theta_C. \quad (12)$$

Because G_W and g_A carry $\sim 10^{-3}$ uncertainties, $\mathcal{K}$ is instead fixed by the measured free-neutron lifetime through $\mathcal{K} = 1/(\tau_n \lambda_0 m_e^5)$, where λ_0 is the dimensionless decay phase-space integral (Phys. Rep. Eqs. 89–91); its value rises from $\bar{\lambda}_0 \simeq 1.636$ (Born) to $\lambda_0^{\text{RC+FM}} \simeq 1.7547$ once corrections are included (App. C). When $T_\nu = T$ the rates obey the detailed-balance relation $\bar{\Gamma}_{p \rightarrow n} = e^{-\Delta/T} \bar{\Gamma}_{n \rightarrow p}$, which the code preserves order by order.

3.3 Corrections beyond Born

Reproducing Y_P to 10^{-3} requires several corrections, applied in sequence (full formulas in App. C):

1. *$T = 0$ radiative corrections*: a Coulomb (Fermi) factor $\mathcal{F}_\pm(E)$ multiplying the integrand when electron and proton are on the same side, times the resummed radiative factor $R(E, k_{\text{max}}) = 1 + \frac{\alpha_{\text{FS}}}{2\pi} \mathcal{C}(E, k_{\text{max}})$ built from Sirlin’s universal function (Phys. Rep. Eqs. 98–101; the Marciano–Sirlin resummation of [2]). Largest single effect, $\Delta Y_P \simeq +0.2\%$.
2. *Bremsstrahlung*: real-photon emission, added to keep detailed balance consistent with the finite- T terms (Phys. Rep. §III.E.2).
3. *Finite-temperature radiative corrections*: stimulated emission/absorption and the electron energy shift in the thermal bath (Brown & Sawyer [4]; Phys. Rep. §III.G), split into three pieces $\Gamma^{\gamma, T} + \Gamma^{\Delta E, T} + \Gamma^{ep+ee, T}$.
4. *Finite nucleon mass*: a Fokker–Planck expansion to first order in T/m_N giving $\chi_\pm^{\text{FM}}$ (Phys. Rep. §III.F; App. C).

5. *Weak magnetism*: at order T/m_N it is equivalent to a shift of the chiral couplings $c_{LL} \rightarrow c_{LL} + f_{\text{wm}} g_A$, $c_{RR} \rightarrow c_{RR} - f_{\text{wm}} g_A$ with $f_{\text{wm}} = (\mu_p - \mu_n)/2 \simeq 1.853$ (Phys. Rep. §III.F; Seckel [5]).
6. *Neutrino spectral distortions*: the equilibrium g_ν is replaced by the actual NEVO neutrino spectrum, which is slightly out of equilibrium.

Figure 4 compares the calibrated rates with H , and Fig. 5 decomposes the total correction to $\Gamma_{n \rightarrow p}$: the $T = 0$ radiative/finite-mass/weak-magnetism block dominates (~ -4 to -6%), while the finite- T and spectral pieces are sub-percent. Figure 6 shows the resulting neutron history.

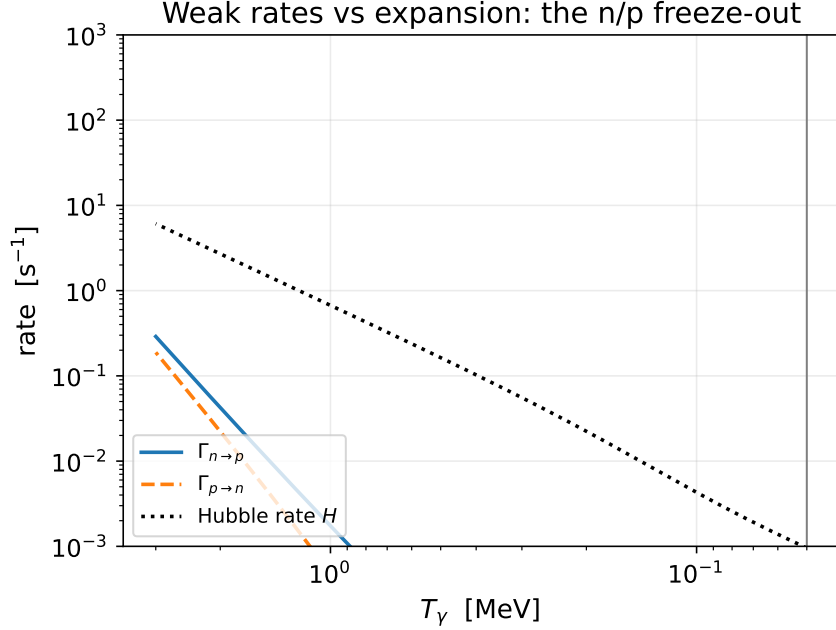


Figure 4: Calibrated weak rates and the Hubble rate H . Equilibrium holds where $\Gamma > H$; the high- T crossing is the n/p freeze-out. (Time flows left to right.)

3.4 In the code

`weak_rates/` evaluates Eq. (11) and all corrections on the background grid. Results are cached in `data/cache_plasma_weak/weak/` behind a configuration *fingerprint*, so a rerun with the same physics reloads them while a changed configuration recomputes. The normalisation $1/(\tau_n \lambda_0 m_e^5)$ is stored as `_NormWeakRates` and the temperature interpolants are `_nT0p_frwrdd/_nT0p_bkwrd` (argument in kelvin).

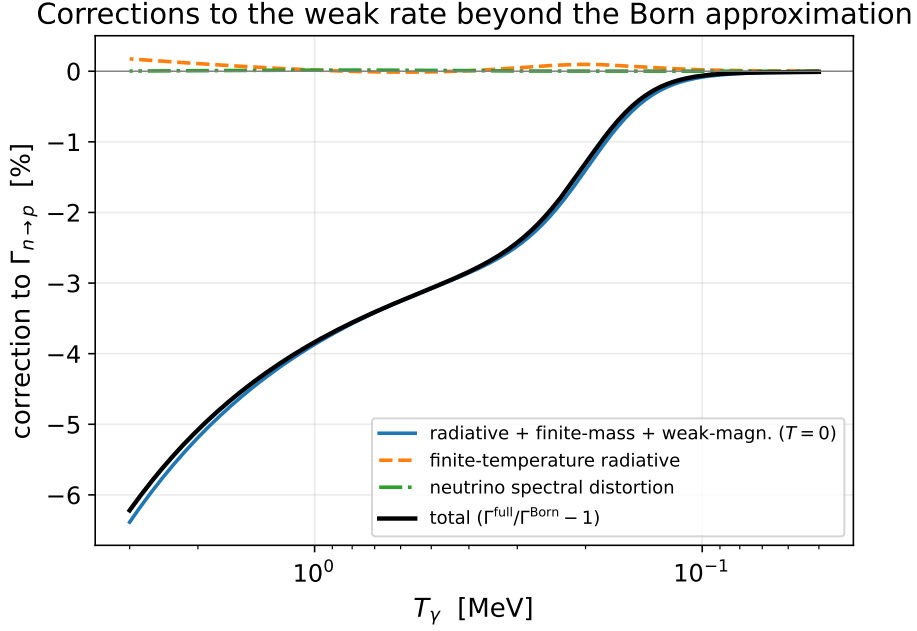


Figure 5: Decomposition of the correction to $\Gamma_{n \rightarrow p}$ relative to the Born rate, obtained by toggling the corresponding flags. The $T = 0$ block (radiative + finite-mass + weak-magnetism) dominates; finite-temperature and spectral-distortion contributions are at the per-mille level.

3.5 Associated options

Option	Default	Effect
<code>tau_n</code>	878.4	Neutron lifetime [s] used to calibrate $\mathcal{K}$.
<code>tau_n_normalization</code>	True	Normalise via τ_n (else from G_F , V_{ud} , g_A).
<code>radiative_corrections</code>	True	$T = 0$ Coulomb + resummed radiative corrections (CCR); False = bare Born rate.
<code>finite_mass_corrections</code>	True	Fokker–Planck finite-nucleon-mass correction.
<code>thermal_corrections</code>	True	Finite-temperature radiative correction (CCRTh; Brown & Sawyer 2001).
<code>spectral_distortions</code>	True	Use the actual NEVO neutrino spectrum in the rates.
<code>sampling_nT0p_per_decade</code>	80	Points per decade of T in the $n \leftrightarrow p$ rate grid.
<code>weak_rate_cache</code>	True	Reuse the fingerprinted <code>data/cache_plasma_weak/weak/</code> cache when valid.

4 Nuclear reactions

4.1 Thermal averaging of reaction rates

In the plasma the nuclei have a Maxwell–Boltzmann distribution of relative velocities (Phys. Rep. Eq. 118), and the thermonuclear rate of a two-body reaction is the thermal average of the cross-section (Phys. Rep. Eq. 119),

$$N_A \langle \sigma v \rangle = N_A \int_0^\infty \sigma(v) \phi_{\text{MB}}(v) v dv, \quad \phi_{\text{MB}}(v) v dv = \sqrt{\frac{8}{\pi m}} \frac{E e^{-E/k_B T}}{(k_B T)^{3/2}} dE, \quad (13)$$

in $\text{cm}^3 \text{s}^{-1} \text{mol}^{-1}$, with m the reduced mass. For charged-particle reactions σ is dominated by Coulomb-barrier tunnelling, so it is written with the astrophysical S -factor $\sigma(E) = S(E) e^{-2\pi\eta}/E$, where $\eta = Z_1 Z_2 e^2 / (\hbar v)$ is the Sommerfeld parameter (Phys. Rep. Eqs. 120–122); $S(E)$ removes most of the energy dependence except for resonances. `primat` does not perform this average at run time: it reads the precomputed $N_A \langle \sigma v \rangle (T_9)$ tables (from the NACRE-II and related evaluations

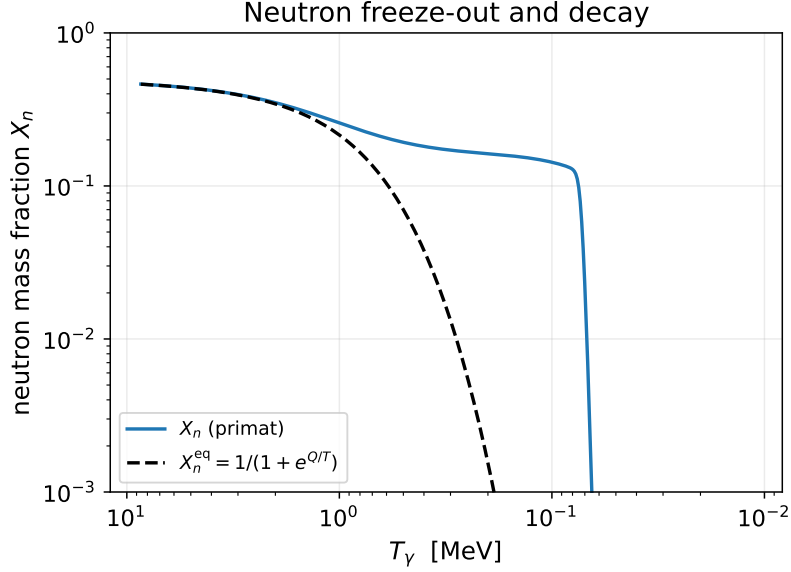


Figure 6: Neutron mass fraction X_n from primat (solid) versus the weak-equilibrium prediction $X_n^{\text{eq}} = 1/(1 + e^{Q/T})$ (dashed). X_n tracks equilibrium until freeze-out near 0.8 MeV, then declines slowly until deuterium formation burns the neutrons into ${}^4\text{He}$.

[8, 9]) and resamples them onto a common T_9 grid (log–log cubic interpolation). Five radiative-capture rates ($n(p, \gamma)d$, $d(p, \gamma){}^3\text{He}$, $t(p, \gamma){}^4\text{He}$, $t(\alpha, \gamma){}^7\text{Li}$, ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$) additionally receive a T_9 -dependent QED correction (`nuclear_qed_corrections`; Pitrou & Pospelov [10]), applied as a multiplicative rescaling of the median rate at load time so that the corrected value becomes the new central rate.

4.2 The reaction network

With $Y_i \equiv n_i/n_b$ (so $\sum_i A_i Y_i = 1$), baryon conservation removes the dilution term and the abundances obey the Wagoner form (Phys. Rep. Eqs. 130–132)

$$\dot{Y}_i = \sum_{\text{reactions}} N_i \left[\Gamma_{j \rightarrow i} \frac{\prod_j Y_j^{N_j}}{\prod_j N_j!} - \Gamma_{i \rightarrow j} \frac{\prod_i Y_i^{N_i}}{\prod_i N_i!} \right], \quad (14)$$

where N_i is the stoichiometric coefficient of species i and the abundance rate is related to the number-density rate by $\Gamma_{i_1 \dots \rightarrow \dots} = n_b^{(\sum N_i)-1} \gamma$ with $\gamma = \langle \sigma v \rangle$. For the common two-body case $i + j \leftrightarrow k + l$ this is

$$\dot{Y}_i \supset Y_k Y_l \Gamma_{kl \rightarrow ij} - Y_i Y_j \Gamma_{ij \rightarrow kl}, \quad \Gamma_{ij \rightarrow kl} = n_b \langle \sigma v \rangle_{ij \rightarrow kl}. \quad (15)$$

Symmetry factors. The factorials $N_i!$ in Eq. (14) are the combinatorial symmetry factors that appear whenever a reaction has *identical* reactants or products. For example $d + d \rightarrow {}^3\text{He} + n$ has two identical deuterons in the entrance channel ($N_d = 2$): the number of distinct reacting pairs in a gas of n_d deuterons is $\frac{1}{2}n_d(n_d - 1) \simeq \frac{1}{2}n_d^2$, so the flux carries a factor $1/2! = 1/2$, and the stoichiometric prefactor $N_d = 2$ means each reaction removes two deuterons. The same $1/N!$ appears for identical products (e.g. ${}^7\text{Li} + p \rightarrow \alpha + \alpha$). Omitting these factors would double-count indistinguishable pairs.

4.3 Reverse rates from detailed balance

Only forward rates are tabulated; reverse rates are fixed by requiring that at nuclear statistical equilibrium (NSE) forward and reverse fluxes cancel. Using the equilibrium number densities

(App. D) the ratio depends only on the nuclide masses, spins and the reaction Q -value, and takes the three-parameter form (Phys. Rep. Eqs. 135–137)

$$\frac{\gamma_{kl \rightarrow ij}}{\gamma_{ij \rightarrow kl}} = \alpha \left(\frac{T}{10^9 \text{ K}} \right)^\beta \exp \left(\frac{\gamma \times 10^9 \text{ K}}{T} \right), \quad (16)$$

with (α, β, γ) computed once per reaction from nuclide data (`compute_detailed_balance_coefficients`). The derivation, including the symmetry factors and the binding-energy exponent, is given in App. D. Figure 7 shows several forward rates together with their reverse (photo-dissociation) rates: the reverse rate rises steeply with T_9 and crosses the forward rate around $T_9 \sim 1$ –2, i.e. where each capture is balanced by its inverse – this is the bottleneck that holds the nuclei at their NSE values until the universe cools enough.

4.4 The three-era ODE integration

The full network is far too stiff to integrate in one shot, so the solve is split at the fixed boundaries $T_{\text{start}}(10 \text{ MeV}) > T_{\text{weak}}(\sim 0.8 \text{ MeV}) > T_{\text{nucl}}(\sim 0.08 \text{ MeV}) > T_{\text{end}}(\sim 10^{-3} \text{ MeV})$, mapped to times through $t(T)$:

- **HT era** ($T_{\text{start}} \rightarrow T_{\text{weak}}$): only n, p , governed by the linear system $\dot{Y}_n = b Y_p - f Y_n$, $\dot{Y}_p = f Y_n - b Y_p$ with $f = \Gamma_{n \rightarrow p}$, $b = \Gamma_{p \rightarrow n}$, started from weak equilibrium $Y_n = b/(b + f)$ and integrated with the non-stiff LSODA solver.
- **MT era** ($T_{\text{weak}} \rightarrow T_{\text{nucl}}$): the fixed 18-reaction subset (even for `large` – the full net is too stiff before the deuterium bottleneck opens), seeded with the HT $\{Y_n, Y_p\}$ and the NSE values Y_i^{NSE} (App. D) for all other species, integrated with the implicit BDF solver and the analytic Jacobian.
- **LT era** ($T_{\text{nucl}} \rightarrow T_{\text{end}}$): the full chosen network (12 / 68 (`large`, `amax=8`) / ~ 429 reactions), seeded from the MT result, BDF + Jacobian.

The normalised $n \leftrightarrow p$ rates feed all three eras as $f, b = \text{NormWR} \times \{\text{_nT0p_frwr}, \text{_nT0p_bkwr}\}$ with $\text{NormWR} = 1/(\tau_n F_n)$ in the `tau_n_normalization` mode (F_n the decay phase-space factor). The reverse nuclear rates of Eq. (16) are clamped at low T_9 so that the exothermic factor e^{γ/T_9} cannot overflow (the “exothermic blow-up” guard in `network_data.py`).

Solvers and the Jacobian. Two of `scipy.integrate.solve_ivp`’s integrators are used. **LSODA** (the Livermore solver with automatic stiffness detection) switches on the fly between a non-stiff Adams–Moulton predictor–corrector and a stiff backward formula; it is used for the smooth, mildly-stiff *background* ODEs $[a(T)]$ of Eq. (5) and $t(a)$ and for the two-species HT system. The MT and LT *nuclear* networks are very stiff – a fast reaction and its detailed-balance reverse differ by many orders of magnitude, so the Jacobian has widely separated eigenvalues – and are integrated with the implicit **BDF** method (backward differentiation formulas), which is unconditionally stable for stiff systems where an explicit method would need prohibitively small steps. A BDF step solves a nonlinear system by Newton iteration and therefore needs the Jacobian $J_{ij} = \partial \dot{Y}_i / \partial Y_j$ of the right-hand side (14). Rather than let the solver estimate J by finite differences (costly and noisy for ~ 60 species), `primat` supplies it analytically: `compile_network` differentiates the polynomial mass-action terms symbolically and emits a fast `njit` kernel (`nucl.JacobianMT/JacobianLT`), passed to `solve_ivp` via its `jac` argument, which both accelerates and stabilises the integration.

4.5 Final observables

At T_{end} the code forms the nine entries of the result dictionary:

$$Y_P^{\text{BBN}} = 4Y_{\text{He}}, \quad \text{D/H} = \frac{Y_{\text{D}}}{Y_p}, \quad \frac{{}^3\text{He}}{\text{H}} = \frac{Y_{3\text{H}} + Y_{3\text{He}}}{Y_p}, \quad \frac{{}^7\text{Li}}{\text{H}} = \frac{Y_{7\text{Li}} + Y_{7\text{Be}}}{Y_p}, \quad (17)$$

where ${}^3\text{H}+{}^3\text{He}$ and ${}^7\text{Be}+{}^7\text{Li}$ include the post-BBN β -decays ${}^3\text{H}\rightarrow{}^3\text{He}$ and ${}^7\text{Be}\rightarrow{}^7\text{Li}$. N_{eff} is read off the neutrino-to-photon energy ratio at T_{end} via Eq. (7).

Y_P^{BBN} **versus** Y_P^{CMB} . $Y_P^{\text{BBN}} = 4Y_{\text{He}}$ is the helium *nucleon* fraction: $Y_{\text{He}} = n_{\text{He}}/n_b$ is the number of ${}^4\text{He}$ per baryon, and multiplying by $A = 4$ gives the fraction of *baryons* (counted as nucleons) locked in ${}^4\text{He}$. This is the natural BBN output but is not exactly a mass fraction, because a bound ${}^4\text{He}$ nucleus is lighter than four free nucleons (binding energy) and the ${}^1\text{H}$ atom mass differs slightly from one atomic mass unit. CMB likelihoods instead parametrise helium by its *true mass fraction* $Y_P = \rho_{\text{He}}/(\rho_{\text{He}} + \rho_{\text{H}})$, so for consistency `primat` also returns

$$Y_P^{\text{CMB}} = \frac{\frac{1}{4} r_{\text{He}} Y_P^{\text{BBN}}}{\frac{1}{4} r_{\text{He}} Y_P^{\text{BBN}} + r_{\text{H}} (1 - Y_P^{\text{BBN}})}, \quad r_{\text{He}} \equiv \frac{m_{\text{He}}}{m_u}, \quad r_{\text{H}} \equiv \frac{m_{\text{H}}}{m_u}, \quad (18)$$

which reweights the nucleon fraction by the actual ${}^4\text{He}$ and ${}^1\text{H}$ masses (in atomic mass units m_u); numerically Y_P^{CMB} is about 0.5% below Y_P^{BBN} . Finally `Omeganurel` / `OneOverOmeganunr` are the relativistic relic-neutrino $\Omega_\nu h^2$ and the inverse non-relativistic $\Omega_\nu h^2$ per summed eV, built from the T_ν/T_γ ratio at T_{end} scaled to today through T_0^{CMB} .

4.6 Networks, eras, and rate variation

The reaction list is named in `data/nuclear/networks/{small,small_parthenope,large}.txt` (or any other network file of the same form, resolved through the `user_nuclear_dir` overlay of §1 if set); `load_network` parses it and `compile_network` turns the stoichiometry of Eq. (14) into fast right-hand-side and Jacobian kernels (the three eras of Sec. 4.4 share these kernels). Setting `amax` filters *any* network (not just `large`) to nuclides of mass number $\leq \text{amax}$, e.g. `network="large", amax=8` reproduces the former “medium” network’s 68 reactions exactly (App. E). Each reaction `<name>` carries a variation knob `p_<name>` sampling its rate at median $\times e^{p\sigma}$ for Monte-Carlo error propagation (§1.2); the additive `delta_<name>` multiplies it by $1 + \delta$ instead, and the combined factor is clamped to `mc_rate_rescale_cap` (default 30, both directions):

```
PRIMAT({"network": "large", "amax": 8,
      "p_d_p_He3_g": 1.0}).solve() # d(p,gamma)He3 at +1 sigma
```

Figure 8 shows the abundance history and Fig. 9 the final abundances versus the baryon density with the observational constraints.

4.7 Rate uncertainties

Each evaluated rate comes with a 1σ uncertainty (a log-normal width σ per reaction). These propagate to the abundances; the dominant ones for Y_P are the neutron lifetime and the $n(p, \gamma)d$, $d(d, n){}^3\text{He}$, $d(d, p)t$ reactions, with the rule-of-thumb (Phys. Rep. Eq. 145)

$$\Delta Y_P \approx 1.5 \times 10^{-3} \frac{\Delta \langle \sigma v \rangle}{\langle \sigma v \rangle}, \quad \Delta Y_P \approx 0.18 \frac{\Delta \tau_n}{\tau_n}, \quad (19)$$

while D/H is most sensitive to $d(p, \gamma){}^3\text{He}$, $d(d, n){}^3\text{He}$ and $d(d, p)t$. In `primat` the full budget is obtained by Monte-Carlo (§1.2), drawing each `p_<rxn>` from $\mathcal{N}(0, 1)$ (and τ_n from its own 1σ); the Python-backend helper `mc_uncertainty` parallelises this with `joblib`, while `primat.backend.run_mc/--mc` dispatch to the multi-threaded C implementation by default. Section 5 quantifies the individual responses.

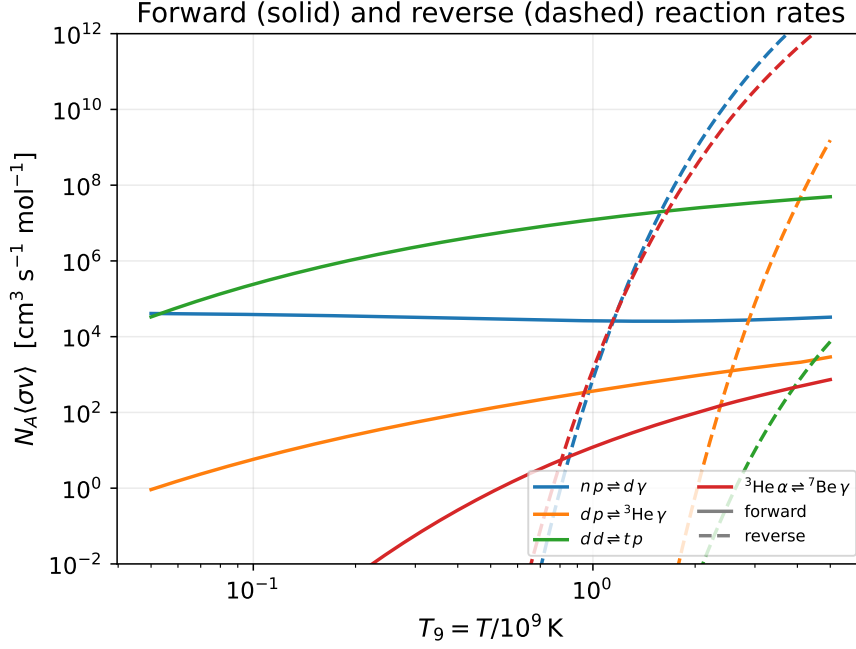


Figure 7: Forward (solid) and reverse (dashed, by detailed balance) rates $N_A\langle\sigma v\rangle(T_9)$ for four key reactions. The forward rates depend only weakly on T_9 ; the reverse photo-dissociation rates rise steeply and cross the forward rates near $T_9 \sim 1-2$.

4.8 Associated options

Option	Default	Effect
network	"small"	Reaction set: small / small_parthenope (12), large (~ 429), or any custom network file.
amax	None	Drop reactions with any nuclide of mass number $> A$, for <i>any</i> network.
nuclear_qed_corrections	True	QED corrections to radiative-capture rates [10].
numerical_precision	1e-7	Relative tolerance of the ODE solver.
rate_grid_npts	1000	Points in the master T_9 grid for rate resampling.
p_<reaction>	0	Log-normal variation of an individual rate ($\times e^{p\sigma}$, MC).
delta_<reaction>	0	Additive fractional variation of an individual rate ($\times(1 + \delta)$).
mc_rate_rescale_cap	30	Clamp on the combined p/δ rate-rescaling factor during MC.
output_time_evolution	False	Populate (and, if <code>output_file</code> is set, write) the full time series.

5 Sensitivity of the abundances

A compact way to understand which inputs matter is the logarithmic sensitivity

$$S(O, p) \equiv \frac{\partial \ln O}{\partial \ln p} \approx \frac{\ln O(p(1 + \delta)) - \ln O(p(1 - \delta))}{2 \ln(1 + \delta)}, \quad (20)$$

the percentage change of observable O per percent change of parameter p , evaluated by a symmetric $\delta = 1\%$ finite difference. Figure 10 shows the matrix for the four light-element observables against the twelve key reaction rates and the physical parameters τ_n , G_N , $\Omega_b h^2$ and ΔN_{eff} . Reading it off: Y_P is almost insensitive to the nuclear rates but responds strongly to ΔN_{eff} (faster expansion $\rightarrow$ earlier freeze-out $\rightarrow$ more neutrons) and to τ_n ; D/H is driven by

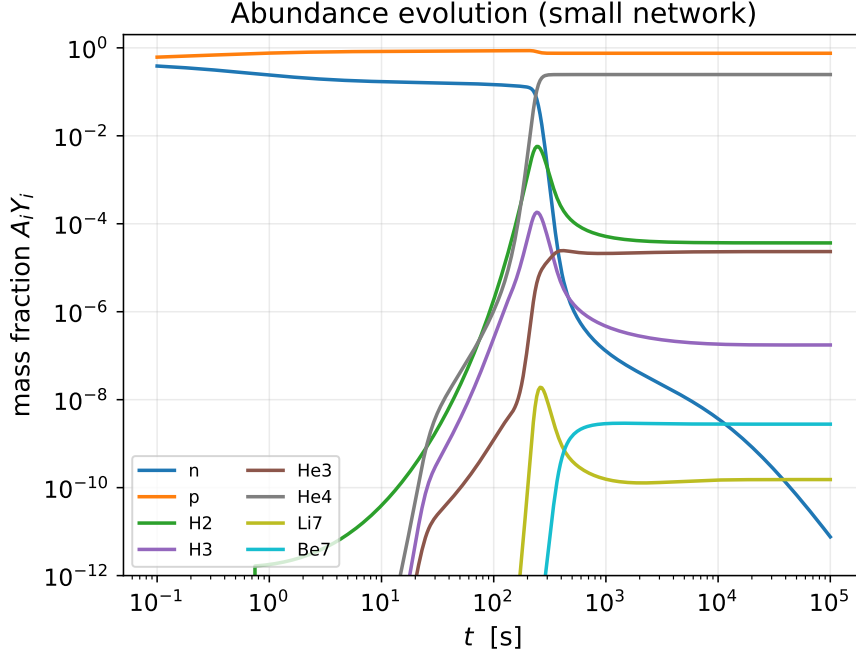


Figure 8: Mass fractions $A_i Y_i(t)$ for the small network. Deuterium builds up and is rapidly burned into ${}^4\text{He}$ once $T \sim 0.07$ MeV (the deuterium bottleneck), leaving the trace primordial D, ${}^3\text{He}$, ${}^7\text{Li}$ and ${}^7\text{Be}$.

$\Omega_b h^2$, G_N and the deuterium-burning rates; ${}^7\text{Li}/\text{H}$ is anti-correlated with $\Omega_b h^2$ along the upper Schramm branch and sensitive to the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ and ${}^7\text{Be}(n, p){}^7\text{Li}$ rates. The figure reproduces `notebooks/Sensitivity.ipynb`.

A Thermodynamic integrals

The Fermi–Dirac (+) and Bose–Einstein (−) integrals used in Eq. (3) are (Phys. Rep. App. A)

$$I_{\pm}^{(n,m)}(x) \equiv \int_0^{\infty} \frac{p^n \mathcal{E}^{m-n}}{e^{\mathcal{E}} \pm 1} dp, \quad \mathcal{E} \equiv \sqrt{p^2 + x^2}, \quad x = m/T, \quad (21)$$

so that $\rho = \frac{gT^4}{2\pi^2} I_{\pm}^{(2,1)}$, $P = \frac{gT^4}{6\pi^2} I_{\pm}^{(0,3)}$ and $n = \frac{gT^3}{2\pi^2} I_{\pm}^{(2,0)}$. Useful limits are $I_+^{(2,1)}(0) = \frac{7\pi^4}{120}$, $I_+^{(0,1)}(0) = \frac{\pi^2}{12}$ and $I_-^{(0,1)} = \frac{\pi^2}{6}$. The electron chemical potential is non-zero only through the tiny charge-neutrality constraint $n_{e^-} - n_{e^+} = n_p$ and is neglected ($\xi_e \ll 1$) in the weak rates; it is kept only where it sets the absolute $e^{\pm}$ number density.

B QED plasma corrections

The thermal $e^{\pm}$ /photon interactions shift the masses by (Phys. Rep. Eqs. 40, 42; [6, 7])

$$\begin{aligned} \frac{\delta m_e^2(p, T)}{T^2} &= \frac{4\alpha_{\text{FS}}}{\pi} \left[I_-^{(0,1)} + I_+^{(0,1)}(x) \right] - \frac{2\bar{m}_e^2 \alpha_{\text{FS}}}{\pi \bar{p}} \int_0^{\infty} d\bar{q} \frac{\bar{q}}{\mathcal{E}_{\bar{q}}} \ln \left| \frac{\bar{p} + \bar{q}}{\bar{p} - \bar{q}} \right| \frac{1}{e^{\mathcal{E}_{\bar{q}}} + 1}, \\ \frac{\delta m_{\gamma}^2}{T^2} &= \frac{8\alpha_{\text{FS}}}{\pi} I_+^{(0,1)}(x), \end{aligned} \quad (22)$$

with $\bar{p} = p/T$, $x = \bar{m}_e = m_e/T$, $\mathcal{E}_{\bar{q}} = \sqrt{\bar{q}^2 + x^2}$. The induced pressure shift is a sum of three terms in increasing order of the electromagnetic coupling e (with $\alpha_{\text{FS}} = e^2/4\pi$).

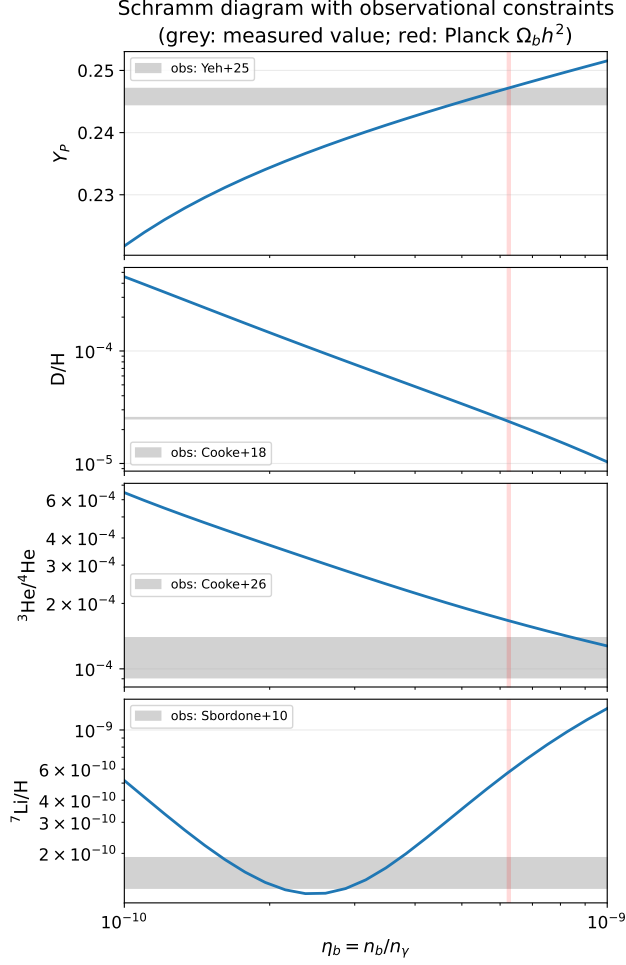


Figure 9: Schramm diagram with observational constraints (grey bands; values from `notebooks/StandardPlots.ipynb`). The red band is the Planck baryon density. Y_P and D/H agree with Planck, ${}^3\text{He}/{}^4\text{He}$ lies somewhat high, and ${}^7\text{Li}/H$ is over-predicted – the “lithium problem”.

`primat` (`qed_pressure.py`) tabulates and *sums the first two*, stored as the two columns of `data/cache_plasma_weak/plasma/QED_*.txt`:

$$\frac{\delta P_a}{T^4} = \frac{\alpha_{\text{FS}}}{\pi} \left[-\frac{2}{3} I_+^{(0,1)}(x) - \frac{2}{\pi^2} \left(I_+^{(0,1)}(x) \right)^2 \right] \quad [O(e^2), \text{ dominant, negative}], \quad (23)$$

$$\frac{\delta P_{e^3}}{T^4} = \frac{4}{3} \sqrt{2\pi} \alpha_{\text{FS}}^{3/2} \left[\frac{I_+^{(0,1)}(x) + I_+^{(2,-1)}(x)}{\pi^2} \right]^{3/2} \quad [O(e^3), \text{ ring/plasmon}], \quad (24)$$

$$\frac{\delta P_b}{T^4} = \frac{\alpha_{\text{FS}}}{\pi^3} \bar{m}_e^2 \iint \frac{\bar{p}^2 d\bar{p} \bar{q}^2 d\bar{q}}{\mathcal{E}_{\bar{p}} \mathcal{E}_{\bar{q}}} \ln \left| \frac{\bar{p} + \bar{q}}{\bar{p} - \bar{q}} \right| \frac{1}{(e^{\mathcal{E}_{\bar{p}}} + 1)(e^{\mathcal{E}_{\bar{q}}} + 1)} \quad [O(e^4), \text{ optional}], \quad (25)$$

with $I_+^{(2,-1)}(x) = \int_0^\infty \sqrt{p^2 + x^2} / (e^{\mathcal{E}} + 1) dp$ (I2m1). δP_a is the one-loop Frenkel–Galitskii–Migdal term, the $O(\alpha_{\text{FS}}^{3/2})$ ring term δP_{e^3} is the Blaizot–Zinn–Justin plasmon resummation (PRIMAT-Main.m), and the two-loop exchange δP_b – equal to the subdominant integral $\overline{\delta P}_s$ of Phys. Rep. Eq. 49 – is *off by default* (it would need a separate flag/file). The energy shift follows from $\rho = -P + T dP/dT$,

$$\overline{\delta \rho} = 3 \overline{\delta P} + \frac{\partial \overline{\delta P}}{\partial \ln T}, \quad \delta g_\rho \equiv \frac{30}{\pi^2} \overline{\delta \rho}, \quad \delta g_P \equiv \frac{90}{\pi^2} \overline{\delta P}, \quad (26)$$

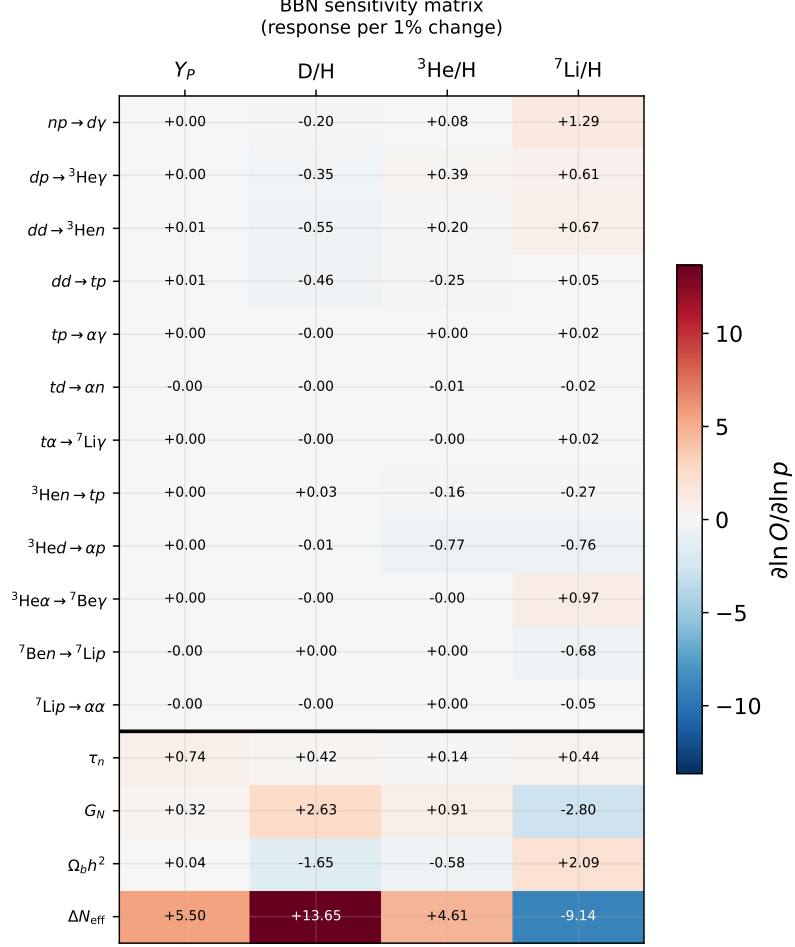


Figure 10: Logarithmic sensitivity $\partial \ln O / \partial \ln p$ of the four observables (columns) to the nuclear rates (top block) and physical parameters (bottom block), per 1% change. Red is positive, blue negative.

with the high- T limit $\delta g_\rho \simeq \delta g_P \simeq -25\alpha_{\text{FS}}/(4\pi)$. These enter the code in four places (Phys. Rep. Eqs. 50–53): (i) the plasma entropy $\mathcal{S}^{\text{QED}} = \mathcal{S} + (\delta g_P + 3\delta g_\rho)/8$ used to obtain $a(T)$; (ii) the neutrino-temperature normalisation $(z_0^{\text{QED}})^3 = \frac{11}{4}(1 - \frac{4}{11}\frac{25\alpha_{\text{FS}}}{8\pi})$, giving $z_0^{\text{QED}} \simeq 1.39979$; (iii) the plasma energy density $\mathcal{E}^{\text{QED}} = \mathcal{E} + \delta g_\rho/2$ in Eq. (4); and (iv) through the electron mass shift – which is exactly the Brown–Sawyer relation $2E\Delta E = \delta m_e^2$ and is therefore counted as a finite-temperature weak-rate correction (App. C).

C Weak rates beyond the Born approximation

Matrix element

With the mostly-minus signature, each rate with nucleon a in the initial and b in the final state is (Phys. Rep. Eq. A40)

$$n_a \Gamma_{a \rightarrow b} = \int \frac{d^3 \mathbf{p}_a d^3 \mathbf{p}_e d^3 \mathbf{p}_\nu}{2^4 (2\pi)^8} \delta(E_a - E_b + s_e E_e + s_\nu E_\nu) \frac{|M|_{a \rightarrow b}^2}{E_n E_p E_e E_\nu} f_a f_\nu(s_\nu E_\nu) f_e(s_e E_e), \quad (27)$$

where $s_{e,\nu} = +1$ (-1) for an initial (final) lepton, and a final-state lepton enters as a Pauli-blocking factor via $f(-E) = 1 - f(E)$. The interaction is $\mathcal{H}_I = \frac{G_F}{\sqrt{2}} J_{e\nu}^\mu J_{pn,\mu}$ with a purely

left-chiral leptonic current and the hadronic current (Phys. Rep. Eqs. A41–A44)

$$J_{pn}^\mu = \cos \theta_C \bar{p} \left(\gamma^\mu (1 - g_A \gamma^5) + i \frac{f_{\text{wm}}}{m_N} 2 \Sigma^{\mu\nu} q_\nu \right) n, \quad f_{\text{wm}} = \frac{\mu_p - \mu_n}{2} \simeq 1.853. \quad (28)$$

Ignoring weak magnetism, the squared matrix element is $\frac{|M|^2}{2^7 G_F^2} = c_{LL} \mathcal{M}_{LL} + c_{RR} \mathcal{M}_{RR} + c_{LR} \mathcal{M}_{LR}$ with

$$c_{LL} = \frac{(1+g_A)^2}{4}, \quad c_{RR} = \frac{(1-g_A)^2}{4}, \quad c_{LR} = \frac{g_A^2 - 1}{4}, \quad (29)$$

and $\mathcal{M}_{LL} = (p_n \cdot p_\nu)(p_p \cdot p_e)$, $\mathcal{M}_{RR} = (p_n \cdot p_e)(p_p \cdot p_\nu)$, $\mathcal{M}_{LR} = m_p m_n (p_\nu \cdot p_e)$ (Phys. Rep. Eq. A46). All six processes of Eq. (9) share this matrix element (crossing + time reversal).

Born rate and calibration

In the infinite-mass limit the three $\mathcal{M}/(E_n E_p E_e E_\nu) \rightarrow 1$ (after an angular average), giving Eqs. (11)–(12). The decay phase-space factor is (Phys. Rep. Eq. 88)

$$\bar{\lambda}_0 = m_e^{-5} \int_0^{\sqrt{\Delta^2 - m_e^2}} p^2 E_\nu^2 dp = \sqrt{\bar{\Delta}^2 - 1} \frac{-8 - 9\bar{\Delta}^2 + 2\bar{\Delta}^4}{60} + \frac{\bar{\Delta}}{4} \text{arccosh } \bar{\Delta} \simeq 1.63609, \quad (30)$$

with $\bar{\Delta} = \Delta/m_e$; including radiative and finite-mass corrections raises it to $\lambda_0^{\text{RC+FM}} \simeq 1.75474$.

$T = 0$ radiative corrections

The corrected forward rate is (Phys. Rep. Eq. 99)

$$\Gamma_{n \rightarrow p}^{\text{RC0}} = \mathcal{K} \int_0^\infty p^2 dp [\mathcal{F}_+(E) R(E, E_\nu) \chi_+(E) + \mathcal{F}_+(-E) R(E, \Delta + E) \chi_+(-E)], \quad (31)$$

where the Fermi (Coulomb) factor $\mathcal{F}_\pm(x) = F(|x|)$ if $\pm x > 0$ and 1 otherwise switches on only when the electron and proton are on the same side. The radiative factor is $R(E, k_{\text{max}}) = 1 + \frac{\alpha_{\text{FS}}}{2\pi} \mathcal{C}(E, k_{\text{max}})$ with Sirlin's function (Phys. Rep. Eqs. A78–A80; [3])

$$\mathcal{C}(E, k_{\text{max}}) = 4 \ln \frac{m_Z}{m_p} + \ln \frac{m_p}{m_A} + 2C + A_g + g(E, k_{\text{max}}), \quad (32)$$

$$g(E, k_{\text{max}}) = 3 \ln \frac{m_p}{m_e} - \frac{3}{4} + \frac{4}{\beta} L \left(\frac{2\beta}{1+\beta} \right) + 4[R(\beta) - 1] \left[\frac{k_{\text{max}}}{3E} - \frac{3}{2} + \ln \frac{2k_{\text{max}}}{m_e} \right] \\ + R(\beta) \left[2(1 + \beta^2) + \frac{k_{\text{max}}^2}{6E^2} - 4\beta R(\beta) \right], \quad (33)$$

with $\beta = p/E$, $R(\beta) = \frac{1}{2\beta} \ln \frac{1+\beta}{1-\beta}$, $L[x] = \int_0^x \frac{\ln(1-t)}{t} dt$ the Spence function, and the constants $C \simeq 0.891$, $A_g \simeq -0.34$, $m_A \simeq 1.2$ GeV. By default the code uses the Marciano–Sirlin resummed form (Phys. Rep. Eq. A81; [2])

$$R(E, k_{\text{max}}) = \left[1 + \frac{\alpha_{\text{FS}}}{2\pi} (g(E, k_{\text{max}}) - 3 \ln \frac{m_p}{2\Delta}) \right] \left(L + \frac{\alpha_{\text{FS}}}{\pi} C + \frac{\alpha_{\text{FS}}}{2\pi} \delta \right) \\ \times \left[S + \frac{\alpha_{\text{FS}}(m_p)}{2\pi} \left(\ln \frac{m_p}{m_A} + A_g \right) + \text{NLL} \right], \quad (34)$$

where $L \simeq 1.02094$ and $S \simeq 1.02248$ are the (resummed) short- and long-distance electroweak factors, $\alpha_{\text{FS}}(m_p) \simeq 1/134$ is the fine-structure constant run to the proton mass, and the two remaining constants are small fixed higher-order pieces: δ is a two-loop/short-distance correction with $\frac{\alpha_{\text{FS}}}{2\pi} \delta \simeq -4.3 \times 10^{-4}$, and $\text{NLL} \simeq -10^{-4}$ is the next-to-leading-log remainder (Phys. Rep. Eq. A81). The reverse rate uses $\chi_+ \rightarrow \chi_-$, $\mathcal{F}_+ \rightarrow \mathcal{F}_-$, preserving detailed balance.

Bremsstrahlung and finite-temperature corrections

To keep detailed balance once the thermal bath is included, the neutron-beta-decay bremsstrahlung is re-added consistently (Phys. Rep. Eq. 100). The finite-temperature radiative correction has three parts (Phys. Rep. Eqs. 104–106; [4]), $\Gamma^T = \Gamma^{\gamma,T} + \Gamma^{\Delta E,T} + \Gamma^{ep+ee,T}$. The photon-exchange part is

$$\Gamma_{n \rightarrow p}^{\gamma,T} = \frac{\alpha_{\text{FS}} \mathcal{K}}{2\pi} \int_0^\infty \frac{dk}{k} \int_{m_e}^\infty dE \left\{ A(E, k) \left[g(-E) \tilde{\chi}_+^A(E, k) + g(E) \tilde{\chi}_+^A(-E, k) \right] \right. \\ \left. - k B(E, k) \left[g(-E) \tilde{\chi}_+^B(E, k) + g(E) \tilde{\chi}_+^B(-E, k) \right] \right\}, \quad (35)$$

with the stimulated-emission/absorption kernels $\tilde{\chi}_\pm^A(E, k) = \tilde{\chi}_\pm(E - k) + \tilde{\chi}_\pm(E + k) - 2\tilde{\chi}_\pm(E)$, $\tilde{\chi}_\pm^B(E, k) = \pm [\tilde{\chi}_\pm(E - k) - \tilde{\chi}_\pm(E + k)]$ (with $\tilde{\chi}_\pm$ the Born kernel without its p^2 measure) and A, B the Bose factors of the photon bath. The electron-energy-shift part is

$$\Gamma_{n \rightarrow p}^{\Delta E,T} = -\frac{2\alpha_{\text{FS}} \mathcal{K}}{\pi} \int dp [\chi_+(E) + \chi_+(-E)] \left[\frac{\pi^2 T^2}{6} + \int_{m_e}^\infty dE' p' g(E') \left(1 - \frac{m_e^2}{E'^2 - E^2} \right) \right], \quad (36)$$

with $E' = \sqrt{p'^2 + m_e^2}$, and the proton–electron + electron self-energy part is (Phys. Rep. Eq. 106)

$$\Gamma_{n \rightarrow p}^{ep+ee,T} = \frac{\alpha_{\text{FS}} \mathcal{K}}{2\pi} \int_{m_e}^\infty dE [\chi_+(E) + \chi_+(-E)] \int_{m_e}^\infty dE' g(E') \frac{E}{E'^2 - E^2} \\ \times \left\{ 8pp' - (E'^2 + E^2) \ln \left(\frac{p + p'}{p - p'} \right)^2 \right. \\ \left. - \frac{E'}{E} (E'^2 + E^2) \ln \left[\frac{(EE' + pp')^2 - m_e^4}{(EE' - pp')^2 - m_e^4} \right] \right\}, \quad (37)$$

all three with $\Gamma_{p \rightarrow n} = \Gamma_{n \rightarrow p}|_{\chi_+ \rightarrow \chi_-}$. These integrals are evaluated with the **vegas** Monte-Carlo library (or **scipy.dblquad**) and cached.

Finite nucleon mass and weak magnetism

The Born rate drops terms E_ν/m_N , E_e/m_N , Δ/m_N , all of order $\epsilon \equiv T/m_N \sim 10^{-3}$ near freeze-out. **primat** keeps them to first order in ϵ (neglecting $O(\epsilon^2) \sim 10^{-6}$) through a Fokker–Planck expansion of Eq. (27) in the energy transferred to the leptons, which adds (Phys. Rep. Eq. 94)

$$\delta\Gamma_{n \rightarrow p}^{\text{FM}} = \mathcal{K} \int_0^\infty p^2 dp [\chi_+^{\text{FM}}(E, g_A) + \chi_+^{\text{FM}}(-E, g_A)], \quad (38)$$

with $\Gamma_{p \rightarrow n}$ obtained by $\chi_+^{\text{FM}}(E, g_A) \rightarrow \chi_-^{\text{FM}}(E, -g_A)$. Because finite-mass and radiative corrections are both percent-level and do not add linearly, the multiplicative radiative factor $\mathcal{F}_\pm(E) R(E, k_{\text{max}})$ is also applied to the finite-mass integrand ($\delta\Gamma^{\text{RC+FM}}$, Phys. Rep. Eq. 95). The kernel $\chi_\pm^{\text{FM}}(E, g_A)$ (Phys. Rep. Eq. A66) is built from the derivatives $g_\nu^{(n,p)}(E_\nu) \equiv \partial_{E_\nu}^p [E_\nu^n g_\nu(E_\nu)]$ and the reduced couplings $\tilde{c}_\pm = \frac{4}{1+3g_A^2} c_\pm$; its $T = 0$ limit (which enters the calibration λ_0) is

$$\chi_+^{\text{FM}}|_{T=0} = c_{LL} \frac{p^2 E_\nu^2}{m_N E} + c_{RR} \frac{E_\nu^3}{m_N} - (c_{LL} + c_{RR} + c_{LR}) \frac{2E_\nu^3 + E_\nu p^2}{m_N} + c_{LR} \frac{p^2 E_\nu^2}{m_N E}. \quad (39)$$

Weak magnetism, at this order, is equivalent to the shift

$$c_{LL} \rightarrow \frac{(1 + g_A)^2}{4} + f_{\text{wm}} g_A, \quad c_{RR} \rightarrow \frac{(1 - g_A)^2}{4} - f_{\text{wm}} g_A, \quad c_{LR} \text{ unchanged}, \quad (40)$$

so it is implemented simply by modifying the couplings entering $\chi_\pm^{\text{FM}}$ (Phys. Rep. §III.F; [5]). Combined with radiative corrections it shifts λ_0 by $\delta\lambda_0^{\text{FM}} \simeq -3.63 \times 10^{-3}$.

Spectral distortions

Finally, the equilibrium g_ν in $\chi_\pm$ is replaced by the actual neutrino occupation $f_\nu = g_\nu + \delta f_\nu$, which deviates from a Fermi–Dirac because of incomplete decoupling (`spectral_distortions=True`). The distortion is small, so the rate is *linearized* in δf_ν : substituting $g_\nu \rightarrow g_\nu + \delta f_\nu$ in $\chi_\pm$ [Eq. (11)] and keeping the first-order term gives the additive correction

$$\delta\Gamma_{n \rightarrow p}^{\text{SD}} = \mathcal{K} \int_0^\infty p^2 dp [\delta\chi_+(E) + \delta\chi_+(-E)], \quad \delta\chi_\pm(E) = (E_\nu^\mp)^2 \delta f_\nu(E_\nu^\mp) g(-E), \quad (41)$$

i.e. the Born kernel with the neutrino Fermi–Dirac replaced by its distortion δf_ν evaluated at the same shifted neutrino energy $E_\nu^\mp = E \mp \Delta$ (and $\chi_+ \rightarrow \chi_-$ for $p \rightarrow n$). `primat` supports two forms of δf_ν (`neutrino_history.py`):

- *NEVO-table* (default, `analytic_distortions=False`): the distortion is read from the full NEVO spectrum file as a function of comoving momentum $y = E_\nu/T_\nu$ and T_γ .
- *Analytic μ/y* (`analytic_distortions=True`; requires instantaneous decoupling), two continuous knobs `delta_xi_nu` and `y_SZ`:

$$\delta f_\nu^\mu(y) = \frac{1}{e^{y-(\xi+\delta\xi)} + 1} - \frac{1}{e^{y-\xi} + 1} \quad (\mu\text{-type, shift of } \xi = \mu_\nu/T_\nu), \quad (42)$$

$$\delta f_\nu^y(y) = f_{\text{FD}}(1 - f_{\text{FD}}) [4 - y(1 - 2f_{\text{FD}})] \quad (y\text{-type / SZ}), \quad (43)$$

where $f_{\text{FD}} = 1/(e^y + 1)$; the y -type shape is $y^{-2} d_y(y^4 d_y f_{\text{FD}})$, the Sunyaev–Zel’dovich form. The same distortion adds an energy density ρ_ν^{SD} to the Friedmann equation (4).

D Detailed balance and nuclear statistical equilibrium

At thermal equilibrium a non-relativistic species has number density (Phys. Rep. Eq. A33)

$$n_i = g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} e^{\xi_i - x_i}, \quad g_i = 2s_i + 1, \quad \xi_i = \mu_i/T, \quad x_i = m_i/T. \quad (44)$$

At NSE a nuclide is in chemical equilibrium with its nucleons, $\mu_i = Z_i \mu_p + (A_i - Z_i) \mu_n$, which gives (Phys. Rep. Eq. A35)

$$n_i^{\text{NSE}} = \frac{g_i m_i^{3/2}}{2^{A_i}} \left(\frac{n_p}{m_p^{3/2}} \right)^{Z_i} \left(\frac{n_n}{m_n^{3/2}} \right)^{A_i - Z_i} \left(\frac{2\pi}{T} \right)^{3(A_i - 1)/2} e^{B_i/T}, \quad B_i = Z_i m_p + (A_i - Z_i) m_n - m_i, \quad (45)$$

with B_i the binding energy. Requiring forward and reverse fluxes to cancel at NSE, the general detailed-balance relation is (Phys. Rep. Eq. 133)

$$\frac{\gamma_{i_1 \dots i_p \rightarrow j_1 \dots j_q}}{\gamma_{j_1 \dots j_q \rightarrow i_1 \dots i_p}} = \frac{N_{i_1}! \dots N_{i_p}!}{N_{j_1}! \dots N_{j_q}!} \left[\frac{n_{j_1}^{N_{j_1}} \dots}{n_{i_1}^{N_{i_1}} \dots} \right]^{\text{NSE}}, \quad (46)$$

where the factorials are exactly the symmetry factors of Eq. (14). Substituting n_i^{NSE} and using that nuclear reactions conserve Z and A (so the n_p, n_n powers cancel) leaves a function of masses, spins and T only (Phys. Rep. Eq. 134),

$$\frac{\gamma_{j \rightarrow i}}{\gamma_{i \rightarrow j}} = \frac{\prod_i \frac{1}{N_i!} [g_i (m_i T / 2\pi)^{3/2}]^{N_i}}{\prod_j \frac{1}{N_j!} [g_j (m_j T / 2\pi)^{3/2}]^{N_j}} \exp\left(\frac{\sum_j m_j - \sum_i m_i}{T}\right), \quad (47)$$

which collapses to the three-constant form of Eq. (16): α gathers the spin/mass/ 2π prefactors, $\beta = \frac{3}{2}(\sum N_i - \sum N_j)$ the power of T , and $\gamma \propto$ the reaction Q -value $\sum_j m_j - \sum_i m_i$. The masses and spins are taken from the AME/NUBASE tables [11]. `primat` computes (α, β, γ) once per reaction in `compute_detailed_balance_coefficients` and stores them in `data/csv/detailed_balance.csv`.

E Nuclear reaction list and sources

PRIMAT’s networks are plain lists of reaction names under `data/nuclear/networks/` (the `small`, `small_parthenope` and `large` files, or any user-supplied network of the same form); each name maps to a rate table under `data/nuclear/tables/` and a source code in `data/csv/reactions_large.csv`’s `ref` column (also readable from each table’s own header line as `ref=...`). Tables 1–3 reproduce every reaction, expanded as $a + b \leftrightarrow c + d$ (each tabulated forward rate is supplemented by its detailed-balance reverse, App. D), with that source code, split into the three tiers of §4.4: the `small` network, the additional reactions that appear at `network="large"`, `amax=8` (the exact reaction set of the former “medium” network), and the remaining reactions of the full `large` network. This appendix is regenerated directly from that data by `manual/generate_tab_reactions.py`, so it is guaranteed to match what `load_network` actually loads for each of the three tiers. Throughout, $a \equiv {}^4\text{He}$ (α), $d \equiv {}^2\text{H}$, $t \equiv {}^3\text{H}$, $g \equiv \gamma$, n, p the nucleons, and $e^\pm$ the leptons of β -decays (one-way, shown with $\rightarrow$). A † marks rates computed from an analytic formula rather than a tabulated grid (hard-coded in PRIMAT-Main.m, including every β -decay); “(borrowed)” marks a rate table copied verbatim from another reaction’s (e.g. a mirror/analogous reaction with no dedicated evaluation).

The source codes follow the convention “AAA YY” = first-author prefix + year (e.g. **And06** = Ando et al. 2006). The principal ones are: **And06** Ando et al. (2006, $n(p, \gamma)d$ EFT); **Mosc0201** Moscoso et al. (2021); **Gom17** Gómez Iñesta, Iliadis & Coc (2017); **Ser04** Serpico et al. (2004); **Ili16** Iliadis et al. (2016); **deSouza19a/b**, **deSouza2020** de Souza, Iliadis & Coc (2019, 2020); **DAACV04** Descouvemont, Adahchour, Angulo, Coc & Vangioni (2004); **NACRE II** Xu et al. (2013) [8]; **NACRE** Angulo et al. (1999); **ILCCF10** Iliadis, Longland, Champagne, Coc & Fitzgerald (2010); **CGXSV12** Coc, Goriely, Xu, Saimpert & Vangioni (2012) [9]; **TALYS2** the TALYS statistical-model code (Koning et al. 2023); **CF88** Caughlan & Fowler (1988); **MF89** Malaney & Fowler (1989); **Rau94** Rauscher et al. (1994); **Wag69** Wagoner (1969); **Aud03** Audi, Wapstra & Thibault (2003, β -decay/mass data); **Nubase** NUBASE2020, Kondev et al. (2021, β -decay half-lives not in Aud03) [12]; **Trezzi2017** Trezzi et al. (LUNA, 2017); **Bar16** Barbagallo et al. (`n_TOF`, 2016); **Rij19** Rijal et al. (2019). The remaining short author-year codes – all carrying the † analytic-fit marker (Bal95, Bar97C, Efr96, FK90, Has09c, Iga95, Kaw91, Koe91, Men12, Nag06, Tang03, Wan91, Wie87/89, TUNL&Cam08, MF89&Hei98, CF88&MF89, Des99Bea01) – decode by the same first-author/year rule, but their full bibliographic entries are not reproduced here; they are catalogued in the compilations of Coc et al. [9] and the NACRE/NACRE-II evaluations [8] and are not all individually retrievable from the shipped tables.

Table 1: Small network (12 reactions). † = analytic fit hard-coded in PRIMAT-Main.m.

#	Reaction	Source	#	Reaction	Source
1	$n + p \leftrightarrow d + \gamma$	And06	7	$t + \alpha \leftrightarrow {}^7\text{Li} + \gamma$	DAACV04
2	$d + p \leftrightarrow {}^3\text{He} + \gamma$	Mosc0201	8	${}^3\text{He} + n \leftrightarrow t + p$	DAACV04
3	$d + d \leftrightarrow {}^3\text{He} + n$	Gom17	9	${}^3\text{He} + d \leftrightarrow \alpha + p$	deSouza19b
4	$d + d \leftrightarrow t + p$	Gom17	10	${}^3\text{He} + \alpha \leftrightarrow {}^7\text{Be} + \gamma$	Ili16
5	$t + p \leftrightarrow \alpha + \gamma$	Ser04	11	${}^7\text{Be} + n \leftrightarrow {}^7\text{Li} + p$	deSouza2020
6	$t + d \leftrightarrow \alpha + n$	deSouza19a	12	${}^7\text{Li} + p \leftrightarrow \alpha + \alpha$	DAACV04

Table 2: **large** network additions at **amax=8** beyond the small network (55 reactions): this is the exact reaction set of the former **medium** network, reproduced by `{network: large, amax: 8}`.

#	Reaction	Source	#	Reaction	Source
1	${}^8\text{B} \rightarrow \alpha + \alpha + e^+$	Aud03 [†]	29	${}^6\text{Li} + n \leftrightarrow t + \alpha$	CF88 [†]
2	${}^8\text{B} + d \leftrightarrow {}^7\text{Be} + {}^3\text{He}$	TALYS2	30	${}^6\text{Li} + p \leftrightarrow {}^7\text{Be} + \gamma$	NACRE II
3	${}^8\text{B} + n \leftrightarrow {}^7\text{Be} + d$	TALYS2	31	${}^6\text{Li} + p \leftrightarrow {}^3\text{He} + \alpha$	NACRE II
4	${}^8\text{B} + n \leftrightarrow {}^6\text{Li} + {}^3\text{He}$	TALYS2	32	${}^6\text{Li} + t \leftrightarrow {}^7\text{Li} + d$	TALYS2
5	${}^8\text{B} + n \leftrightarrow \alpha + \alpha + p$	TALYS2	33	${}^6\text{Li} + t \leftrightarrow {}^8\text{Li} + p$	TALYS2
6	${}^8\text{B} + t \leftrightarrow {}^7\text{Be} + \alpha$	TALYS2	34	${}^6\text{Li} + t \leftrightarrow \alpha + \alpha + n$	TALYS2
7	${}^8\text{B} + t \leftrightarrow \alpha + \alpha + {}^3\text{He}$	TALYS2	35	${}^7\text{Li} + {}^3\text{He} \leftrightarrow {}^6\text{Li} + \alpha$	TALYS2
8	${}^7\text{Be} + {}^3\text{He} \leftrightarrow \alpha + \alpha + p + p$	CF88&MF89 [†]	36	${}^7\text{Li} + {}^3\text{He} \leftrightarrow \alpha + \alpha + d$	TALYS2
9	${}^7\text{Be} + {}^3\text{He} \leftrightarrow p + p + \alpha + \alpha$	TALYS2	37	${}^7\text{Li} + {}^3\text{He} \leftrightarrow \alpha + \alpha + n + p$	CF88&MF89 [†]
10	${}^7\text{Be} \rightarrow {}^7\text{Li} + e^+$	Aud03 [†]	38	${}^7\text{Li} + d \leftrightarrow {}^8\text{Li} + p$	MF89 [†]
11	${}^7\text{Be} + d \leftrightarrow \alpha + \alpha + p$	Rij19	39	${}^7\text{Li} + d \leftrightarrow \alpha + \alpha + n$	CGXSV12
12	${}^7\text{Be} + n \leftrightarrow \alpha + \alpha$	Bar16	40	${}^7\text{Li} + n \leftrightarrow {}^8\text{Li} + \gamma$	MF89&Hei98 [†]
13	${}^7\text{Be} + p \leftrightarrow {}^8\text{B} + \gamma$	NACRE II	41	${}^7\text{Li} + p \leftrightarrow \alpha + \alpha + \gamma$	NACRE II
14	${}^7\text{Be} + t \leftrightarrow {}^6\text{Li} + \alpha$	TALYS2	42	${}^7\text{Li} + t \leftrightarrow \alpha + \alpha + n + n$	CF88&MF89 [†]
15	${}^7\text{Be} + t \leftrightarrow {}^7\text{Li} + {}^3\text{He}$	TALYS2	43	${}^8\text{Li} + {}^3\text{He} \leftrightarrow {}^7\text{Li} + \alpha$	TALYS2
16	${}^7\text{Be} + t \leftrightarrow \alpha + \alpha + d$	TALYS2	44	${}^8\text{Li} + {}^3\text{He} \leftrightarrow \alpha + \alpha + t$	TALYS2
17	${}^7\text{Be} + t \leftrightarrow \alpha + \alpha + n + p$	CF88&MF89 [†]	45	${}^8\text{Li} \rightarrow \alpha + \alpha + e^-$	Aud03 [†]
18	${}^3\text{He} + {}^3\text{He} \leftrightarrow \alpha + p + p$	NACRE II	46	${}^8\text{Li} + d \leftrightarrow {}^7\text{Li} + t$	Has09c [†]
19	${}^3\text{He} + n \leftrightarrow \alpha + \gamma$	Wag69 [†]	47	${}^8\text{Li} + p \leftrightarrow \alpha + \alpha + n$	Men12 [†]
20	${}^3\text{He} + t \leftrightarrow {}^6\text{Li} + \gamma$	FK90 [†]	48	$\alpha + n + n \leftrightarrow {}^6\text{He} + \gamma$	Efr96 [†]
21	${}^3\text{He} + t \leftrightarrow \alpha + d$	CF88 [†]	49	$\alpha + n + p \leftrightarrow {}^6\text{Li} + \gamma$	CF88 [†]
22	${}^3\text{He} + t \leftrightarrow \alpha + n + p$	CF88 [†]	50	$d + \alpha \leftrightarrow {}^6\text{Li} + \gamma$	Trezzi2017
23	${}^6\text{He} \rightarrow {}^6\text{Li} + e^-$	Aud03 [†]	51	$d + d \leftrightarrow \alpha + \gamma$	NACRE II
24	${}^6\text{Li} + {}^3\text{He} \leftrightarrow {}^7\text{Be} + d$	TALYS2	52	$d + n \leftrightarrow t + \gamma$	Nag06 [†]
25	${}^6\text{Li} + {}^3\text{He} \leftrightarrow \alpha + \alpha + p$	TALYS2	53	$p + p + n \leftrightarrow d + p$	CF88 [†]
26	${}^6\text{Li} + d \leftrightarrow {}^7\text{Be} + n$	MF89 [†]	54	$t \rightarrow {}^3\text{He} + e^-$	Aud03 [†]
27	${}^6\text{Li} + d \leftrightarrow {}^7\text{Li} + p$	MF89 [†]	55	$t + t \leftrightarrow \alpha + n + n$	Nag06 [†]
28	${}^6\text{Li} + n \leftrightarrow {}^7\text{Li} + \gamma$	MF89 [†]			

Table 3: Full **large** network additions beyond **amax=8** (361 reactions).

#	Reaction	Source	#	Reaction	Source
1	${}^{10}\text{B} + {}^3\text{He} \leftrightarrow {}^{11}\text{C} + d$	TALYS2	182	${}^{13}\text{C} + {}^3\text{He} \leftrightarrow {}^{15}\text{N} + p$	TALYS2
2	${}^{10}\text{B} + {}^3\text{He} \leftrightarrow {}^{12}\text{C} + p$	TALYS2	183	${}^{13}\text{C} + {}^3\text{He} \leftrightarrow {}^{15}\text{O} + n$	TALYS2
3	${}^{10}\text{B} + {}^3\text{He} \leftrightarrow {}^{12}\text{N} + n$	TALYS2	184	${}^{13}\text{C} + {}^3\text{He} \leftrightarrow {}^{16}\text{O} + \gamma$	TALYS2
4	${}^{10}\text{B} + {}^3\text{He} \leftrightarrow {}^{13}\text{N} + \gamma$	TALYS2	185	${}^{13}\text{C} + \alpha \leftrightarrow {}^{16}\text{O} + n$	NACRE II
5	${}^{10}\text{B} + \alpha \leftrightarrow {}^{12}\text{C} + d$	TALYS2	186	${}^{13}\text{C} + \alpha \leftrightarrow {}^{17}\text{O} + \gamma$	TALYS2
6	${}^{10}\text{B} + \alpha \leftrightarrow {}^{13}\text{C} + p$	TALYS2	187	${}^{13}\text{C} + d \leftrightarrow {}^{11}\text{B} + \alpha$	TALYS2
7	${}^{10}\text{B} + \alpha \leftrightarrow {}^{13}\text{N} + n$	CF88 [†]	188	${}^{13}\text{C} + d \leftrightarrow {}^{14}\text{C} + p$	TALYS2
8	${}^{10}\text{B} + \alpha \leftrightarrow {}^{14}\text{N} + \gamma$	TALYS2	189	${}^{13}\text{C} + d \leftrightarrow {}^{14}\text{N} + n$	TALYS2
9	${}^{10}\text{B} + d \leftrightarrow {}^{11}\text{B} + p$	TALYS2	190	${}^{13}\text{C} + d \leftrightarrow {}^{15}\text{N} + \gamma$	TALYS2
10	${}^{10}\text{B} + d \leftrightarrow {}^{11}\text{C} + n$	TALYS2	191	${}^{13}\text{C} + n \leftrightarrow {}^{14}\text{C} + \gamma$	TALYS2
11	${}^{10}\text{B} + d \leftrightarrow {}^{12}\text{C} + \gamma$	TALYS2	192	${}^{13}\text{C} + p \leftrightarrow {}^{14}\text{N} + \gamma$	NACRE II
12	${}^{10}\text{B} + d \leftrightarrow \alpha + \alpha + \alpha$	TALYS2	193	${}^{13}\text{C} + t \leftrightarrow {}^{12}\text{B} + \alpha$	TALYS2
13	${}^{10}\text{B} + n \leftrightarrow {}^{11}\text{B} + \gamma$	TALYS2	194	${}^{13}\text{C} + t \leftrightarrow {}^{15}\text{C} + p$	TALYS2
14	${}^{10}\text{B} + n \leftrightarrow {}^{10}\text{Be} + p$	TALYS2	195	${}^{13}\text{C} + t \leftrightarrow {}^{15}\text{N} + n$	TALYS2
15	${}^{10}\text{B} + n \leftrightarrow \alpha + \alpha + t$	TALYS2	196	${}^{13}\text{C} + t \leftrightarrow {}^{16}\text{N} + \gamma$	TALYS2
16	${}^{10}\text{B} + p \leftrightarrow {}^7\text{Be} + \alpha$	NACRE II	197	${}^{14}\text{C} + {}^3\text{He} \leftrightarrow {}^{13}\text{C} + \alpha$	TALYS2
17	${}^{10}\text{B} + p \leftrightarrow {}^{11}\text{C} + \gamma$	NACRE II	198	${}^{14}\text{C} + {}^3\text{He} \leftrightarrow {}^{16}\text{N} + p$	TALYS2
18	${}^{10}\text{B} + p \leftrightarrow \alpha + \alpha + {}^3\text{He}$	TALYS2	199	${}^{14}\text{C} + {}^3\text{He} \leftrightarrow {}^{16}\text{O} + n$	TALYS2
19	${}^{10}\text{B} + t \leftrightarrow {}^{11}\text{B} + d$	TALYS2	200	${}^{14}\text{C} + {}^3\text{He} \leftrightarrow {}^{17}\text{O} + \gamma$	TALYS2
20	${}^{10}\text{B} + t \leftrightarrow {}^{12}\text{B} + p$	TALYS2	201	${}^{14}\text{C} \rightarrow {}^{14}\text{N} + e^-$	Nubase [†]
21	${}^{10}\text{B} + t \leftrightarrow {}^9\text{Be} + \alpha$	TALYS2	202	${}^{14}\text{C} + \alpha \leftrightarrow {}^{18}\text{O} + \gamma$	ILCCF10
22	${}^{10}\text{B} + t \leftrightarrow {}^{12}\text{C} + n$	TALYS2	203	${}^{14}\text{C} + d \leftrightarrow {}^{12}\text{B} + \alpha$	TALYS2
23	${}^{10}\text{B} + t \leftrightarrow {}^{13}\text{C} + \gamma$	TALYS2	204	${}^{14}\text{C} + d \leftrightarrow {}^{15}\text{N} + n$	Kaw91 [†]
24	${}^{11}\text{B} + {}^3\text{He} \leftrightarrow {}^{10}\text{B} + \alpha$	TALYS2	205	${}^{14}\text{C} + d \leftrightarrow {}^{16}\text{N} + \gamma$	TALYS2
25	${}^{11}\text{B} + {}^3\text{He} \leftrightarrow {}^{12}\text{C} + d$	TALYS2	206	${}^{14}\text{C} + n \leftrightarrow {}^{15}\text{C} + \gamma$	Kaw91 [†]
26	${}^{11}\text{B} + {}^3\text{He} \leftrightarrow {}^{13}\text{C} + p$	TALYS2	207	${}^{14}\text{C} + p \leftrightarrow {}^{15}\text{N} + \gamma$	ILCCF10
27	${}^{11}\text{B} + {}^3\text{He} \leftrightarrow {}^{13}\text{N} + n$	TALYS2	208	${}^{14}\text{C} + t \leftrightarrow {}^{16}\text{N} + n$	TALYS2
28	${}^{11}\text{B} + {}^3\text{He} \leftrightarrow {}^{14}\text{N} + \gamma$	TALYS2	209	${}^{14}\text{C} + t \leftrightarrow {}^{17}\text{N} + \gamma$	TALYS2

table 3 continued

#	Reaction	Source	#	Reaction	Source
29	$^{11}\text{B} + \alpha \leftrightarrow ^{14}\text{C} + p$	Wan91 [†]	210	$^{15}\text{C} \rightarrow ^{15}\text{N} + e^-$	Aud03 [†]
30	$^{11}\text{B} + \alpha \leftrightarrow ^{14}\text{N} + n$	NACRE II	211	$^{15}\text{C} + \alpha \leftrightarrow ^{18}\text{O} + n$	TALYS2
31	$^{11}\text{B} + \alpha \leftrightarrow ^{15}\text{N} + \gamma$	Wan91 [†]	212	$^{15}\text{C} + \alpha \leftrightarrow ^{19}\text{O} + \gamma$	TALYS2
32	$^{11}\text{B} + d \leftrightarrow ^{12}\text{B} + p$	CGXSV12	213	$^{15}\text{C} + n \leftrightarrow ^{16}\text{C} + \gamma$	TALYS2
33	$^{11}\text{B} + d \leftrightarrow ^9\text{Be} + \alpha$	TALYS2	214	$^{15}\text{C} + p \leftrightarrow ^{12}\text{B} + \alpha$	TALYS2
34	$^{11}\text{B} + d \leftrightarrow ^{12}\text{C} + n$	CGXSV12	215	$^{15}\text{C} + p \leftrightarrow ^{14}\text{C} + d$	TALYS2
35	$^{11}\text{B} + d \leftrightarrow ^{13}\text{C} + \gamma$	TALYS2	216	$^{15}\text{C} + p \leftrightarrow ^{15}\text{N} + n$	TALYS2
36	$^{11}\text{B} + n \leftrightarrow ^{12}\text{B} + \gamma$	CGXSV12	217	$^{15}\text{C} + p \leftrightarrow ^{16}\text{N} + \gamma$	TALYS2
37	$^{11}\text{B} + p \leftrightarrow ^{11}\text{C} + n$	NACRE [†]	218	$^{16}\text{C} \rightarrow ^{16}\text{N} + e^-$	Aud03 [†]
38	$^{11}\text{B} + p \leftrightarrow ^{12}\text{C} + \gamma$	NACRE II	219	$^9\text{C} \rightarrow \alpha + \alpha + p + e^+$	Aud03 [†]
39	$^{11}\text{B} + p \leftrightarrow \alpha + \alpha + \alpha$	NACRE II	220	$^9\text{C} + \alpha \leftrightarrow ^{12}\text{N} + p$	Wie89 [†]
40	$^{11}\text{B} + t \leftrightarrow ^{13}\text{B} + p$	TALYS2	221	$^9\text{C} + \alpha \leftrightarrow ^{13}\text{O} + \gamma$	TALYS2
41	$^{11}\text{B} + t \leftrightarrow ^{10}\text{Be} + \alpha$	TALYS2	222	$^9\text{C} + d \leftrightarrow ^{10}\text{C} + p$	TALYS2
42	$^{11}\text{B} + t \leftrightarrow ^{13}\text{C} + n$	TALYS2	223	$^9\text{C} + n \leftrightarrow ^8\text{B} + d$	TALYS2
43	$^{11}\text{B} + t \leftrightarrow ^{14}\text{C} + \gamma$	TALYS2	224	$^9\text{C} + n \leftrightarrow ^7\text{Be} + ^3\text{He}$	TALYS2
44	$^{12}\text{B} + ^3\text{He} \leftrightarrow ^{11}\text{B} + \alpha$	TALYS2	225	$^9\text{C} + n \leftrightarrow ^{10}\text{C} + \gamma$	TALYS2
45	$^{12}\text{B} + ^3\text{He} \leftrightarrow ^{14}\text{C} + p$	TALYS2	226	$^9\text{C} + t \leftrightarrow ^8\text{B} + \alpha$	TALYS2
46	$^{12}\text{B} + ^3\text{He} \leftrightarrow ^{14}\text{N} + n$	TALYS2	227	$^9\text{C} + t \leftrightarrow ^{11}\text{C} + p$	TALYS2
47	$^{12}\text{B} + ^3\text{He} \leftrightarrow ^{15}\text{N} + \gamma$	TALYS2	228	$^9\text{C} + t \leftrightarrow ^{12}\text{N} + \gamma$	TALYS2
48	$^{12}\text{B} \rightarrow ^{12}\text{C} + e^-$	Aud03 [†]	229	$^{17}\text{F} \rightarrow ^{17}\text{O} + e^+$	Aud03 [†]
49	$^{12}\text{B} + \alpha \leftrightarrow ^{15}\text{N} + n$	TALYS2	230	$^{17}\text{F} + \alpha \leftrightarrow ^{21}\text{Na} + \gamma$	TALYS2
50	$^{12}\text{B} + \alpha \leftrightarrow ^{16}\text{N} + \gamma$	TALYS2	231	$^{17}\text{F} + \alpha \leftrightarrow ^{20}\text{Ne} + p$	TALYS2
51	$^{12}\text{B} + d \leftrightarrow ^{13}\text{B} + p$	TALYS2	232	$^{17}\text{F} + n \leftrightarrow ^{18}\text{F} + \gamma$	TALYS2
52	$^{12}\text{B} + d \leftrightarrow ^{10}\text{Be} + \alpha$	TALYS2	233	$^{17}\text{F} + n \leftrightarrow ^{14}\text{N} + \alpha$	NACRE [†]
53	$^{12}\text{B} + d \leftrightarrow ^{13}\text{C} + n$	TALYS2	234	$^{17}\text{F} + n \leftrightarrow ^{17}\text{O} + p$	TALYS2
54	$^{12}\text{B} + d \leftrightarrow ^{14}\text{C} + \gamma$	TALYS2	235	$^{17}\text{F} + p \leftrightarrow ^{18}\text{Ne} + \gamma$	ILCCF10
55	$^{12}\text{B} + n \leftrightarrow ^{13}\text{B} + \gamma$	TALYS2	236	$^{18}\text{F} \rightarrow ^{18}\text{O} + e^+$	Aud03 [†]
56	$^{12}\text{B} + p \leftrightarrow ^9\text{Be} + \alpha$	TALYS2	237	$^{18}\text{F} + \alpha \leftrightarrow ^{22}\text{Na} + \gamma$	TALYS2
57	$^{12}\text{B} + p \leftrightarrow ^{12}\text{C} + n$	TALYS2	238	$^{18}\text{F} + \alpha \leftrightarrow ^{21}\text{Ne} + p$	TALYS2
58	$^{12}\text{B} + p \leftrightarrow ^{13}\text{C} + \gamma$	TALYS2	239	$^{18}\text{F} + n \leftrightarrow ^{19}\text{F} + \gamma$	TALYS2
59	$^{12}\text{B} + t \leftrightarrow ^{11}\text{Be} + \alpha$	TALYS2	240	$^{18}\text{F} + n \leftrightarrow ^{15}\text{N} + \alpha$	CF88 [†]
60	$^{12}\text{B} + t \leftrightarrow ^{14}\text{C} + n$	TALYS2	241	$^{18}\text{F} + n \leftrightarrow ^{18}\text{O} + p$	TALYS2
61	$^{12}\text{B} + t \leftrightarrow ^{15}\text{C} + \gamma$	TALYS2	242	$^{18}\text{F} + p \leftrightarrow ^{19}\text{Ne} + \gamma$	ILCCF10
62	$^{13}\text{B} \rightarrow ^{13}\text{C} + e^-$	Aud03 [†]	243	$^{18}\text{F} + p \leftrightarrow ^{15}\text{O} + \alpha$	ILCCF10
63	$^{14}\text{B} \rightarrow ^{14}\text{C} + e^-$	Aud03 [†]	244	$^{19}\text{F} + \alpha \leftrightarrow ^{23}\text{Na} + \gamma$	TALYS2
64	$^{15}\text{B} \rightarrow ^{15}\text{C} + e^-$	Aud03 [†]	245	$^{19}\text{F} + \alpha \leftrightarrow ^{22}\text{Ne} + p$	TALYS2
65	$^8\text{B} + ^3\text{He} \leftrightarrow ^{10}\text{C} + p$	TALYS2	246	$^{19}\text{F} + n \leftrightarrow ^{20}\text{F} + \gamma$	TALYS2
66	$^8\text{B} + \alpha \leftrightarrow ^{11}\text{C} + p$	TALYS2	247	$^{19}\text{F} + p \leftrightarrow ^{20}\text{Ne} + \gamma$	NACRE
67	$^8\text{B} + \alpha \leftrightarrow ^{12}\text{N} + \gamma$	TALYS2	248	$^{19}\text{F} + p \leftrightarrow ^{16}\text{O} + \alpha$	NACRE
68	$^8\text{B} + d \leftrightarrow ^{10}\text{C} + \gamma$	TALYS2	249	$^{20}\text{F} \rightarrow ^{20}\text{Ne} + e^-$	Aud03 [†]
69	$^8\text{B} + p \leftrightarrow ^9\text{C} + \gamma$	Des99Bea01 [†]	250	$^6\text{Li} + \alpha \leftrightarrow ^{10}\text{B} + \gamma$	CF88 [†]
70	$^8\text{B} + t \leftrightarrow ^{10}\text{B} + p$	TALYS2	251	$^6\text{Li} + t \leftrightarrow ^9\text{Be} + \gamma$	TALYS2
71	$^8\text{B} + t \leftrightarrow ^{10}\text{C} + n$	TALYS2	252	$^7\text{Li} + ^3\text{He} \leftrightarrow ^{10}\text{B} + \gamma$	TALYS2
72	$^8\text{B} + t \leftrightarrow ^{11}\text{C} + \gamma$	TALYS2	253	$^7\text{Li} + ^3\text{He} \leftrightarrow ^9\text{Be} + p$	TALYS2
73	$^{10}\text{Be} + ^3\text{He} \leftrightarrow ^{10}\text{B} + t$	TALYS2	254	$^7\text{Li} + \alpha \leftrightarrow ^{10}\text{B} + n$	NACRE [†]
74	$^{10}\text{Be} + ^3\text{He} \leftrightarrow ^{11}\text{B} + d$	TALYS2	255	$^7\text{Li} + \alpha \leftrightarrow ^{11}\text{B} + \gamma$	NACRE II
75	$^{10}\text{Be} + ^3\text{He} \leftrightarrow ^{12}\text{B} + p$	TALYS2	256	$^7\text{Li} + d \leftrightarrow ^9\text{Be} + \gamma$	CGXSV12
76	$^{10}\text{Be} + ^3\text{He} \leftrightarrow ^9\text{Be} + \alpha$	TALYS2	257	$^7\text{Li} + t \leftrightarrow ^{10}\text{Be} + \gamma$	TALYS2
77	$^{10}\text{Be} + ^3\text{He} \leftrightarrow ^{12}\text{C} + n$	TALYS2	258	$^7\text{Li} + t \leftrightarrow ^9\text{Be} + n$	CGXSV12
78	$^{10}\text{Be} + ^3\text{He} \leftrightarrow ^{13}\text{C} + \gamma$	TALYS2	259	$^8\text{Li} + ^3\text{He} \leftrightarrow ^{10}\text{B} + n$	TALYS2
79	$^{10}\text{Be} \rightarrow ^{10}\text{B} + e^-$	Nubase [†]	260	$^8\text{Li} + ^3\text{He} \leftrightarrow ^{11}\text{B} + \gamma$	TALYS2
80	$^{10}\text{Be} + \alpha \leftrightarrow ^{13}\text{C} + n$	TALYS2	261	$^8\text{Li} + ^3\text{He} \leftrightarrow ^{10}\text{Be} + p$	TALYS2
81	$^{10}\text{Be} + \alpha \leftrightarrow ^{14}\text{C} + \gamma$	TALYS2	262	$^8\text{Li} + ^3\text{He} \leftrightarrow ^9\text{Be} + d$	TALYS2
82	$^{10}\text{Be} + d \leftrightarrow ^{11}\text{B} + n$	TALYS2	263	$^8\text{Li} + \alpha \leftrightarrow ^{11}\text{B} + n$	CGXSV12
83	$^{10}\text{Be} + d \leftrightarrow ^{12}\text{B} + \gamma$	TALYS2	264	$^8\text{Li} + \alpha \leftrightarrow ^{12}\text{B} + \gamma$	TALYS2
84	$^{10}\text{Be} + d \leftrightarrow ^8\text{Li} + \alpha$	TALYS2	265	$^8\text{Li} + d \leftrightarrow ^{10}\text{Be} + \gamma$	TALYS2
85	$^{10}\text{Be} + n \leftrightarrow ^{11}\text{Be} + \gamma$	Rau94 [†]	266	$^8\text{Li} + d \leftrightarrow ^9\text{Be} + n$	Bal95 [†]
86	$^{10}\text{Be} + p \leftrightarrow ^{11}\text{B} + \gamma$	TALYS2	267	$^8\text{Li} + d \leftrightarrow ^9\text{Li} + p$	Bal95 [†]
87	$^{10}\text{Be} + p \leftrightarrow ^7\text{Li} + \alpha$	TALYS2	268	$^8\text{Li} + n \leftrightarrow ^9\text{Li} + \gamma$	Rau94 [†]
88	$^{10}\text{Be} + p \leftrightarrow \alpha + \alpha + t$	TALYS2	269	$^8\text{Li} + p \leftrightarrow ^9\text{Be} + \gamma$	TUNL&Cam08 [†]
89	$^{10}\text{Be} + t \leftrightarrow ^{12}\text{B} + n$	TALYS2	270	$^8\text{Li} + t \leftrightarrow ^{10}\text{Be} + n$	TALYS2
90	$^{10}\text{Be} + t \leftrightarrow ^{13}\text{B} + \gamma$	TALYS2	271	$^8\text{Li} + t \leftrightarrow ^{11}\text{Be} + \gamma$	TALYS2
91	$^{10}\text{Be} + t \leftrightarrow ^9\text{Li} + \alpha$	TALYS2	272	$^9\text{Li} + ^3\text{He} \leftrightarrow ^{11}\text{B} + n$	TALYS2
92	$^{11}\text{Be} + ^3\text{He} \leftrightarrow ^{13}\text{B} + p$	TALYS2	273	$^9\text{Li} + ^3\text{He} \leftrightarrow ^{12}\text{B} + \gamma$	TALYS2
93	$^{11}\text{Be} + ^3\text{He} \leftrightarrow ^{10}\text{Be} + \alpha$	TALYS2	274	$^9\text{Li} + ^3\text{He} \leftrightarrow ^{10}\text{Be} + d$	TALYS2
94	$^{11}\text{Be} + ^3\text{He} \leftrightarrow ^{13}\text{C} + n$	TALYS2	275	$^9\text{Li} + ^3\text{He} \leftrightarrow ^{11}\text{Be} + p$	TALYS2
95	$^{11}\text{Be} + ^3\text{He} \leftrightarrow ^{14}\text{C} + \gamma$	TALYS2	276	$^9\text{Li} + ^3\text{He} \leftrightarrow ^9\text{Be} + t$	TALYS2
96	$^{11}\text{Be} \rightarrow ^{11}\text{B} + e^-$	Aud03 [†]	277	$^9\text{Li} + ^3\text{He} \leftrightarrow ^8\text{Li} + \alpha$	TALYS2
97	$^{11}\text{Be} + \alpha \leftrightarrow ^{14}\text{C} + n$	TALYS2	278	$^9\text{Li} \rightarrow ^9\text{Be} + e^-$	Aud03 [†]
98	$^{11}\text{Be} + \alpha \leftrightarrow ^{15}\text{C} + \gamma$	TALYS2	279	$^9\text{Li} \rightarrow \alpha + \alpha + n + e^-$	Aud03 [†]

table 3 continued

#	Reaction	Source	#	Reaction	Source
99	$^{11}\text{Be} + d \leftrightarrow ^{12}\text{B} + n$	TALYS2	280	$^9\text{Li} + \alpha \leftrightarrow ^{12}\text{B} + n$	CGXSV12
100	$^{11}\text{Be} + d \leftrightarrow ^{13}\text{B} + \gamma$	TALYS2	281	$^9\text{Li} + \alpha \leftrightarrow ^{13}\text{B} + \gamma$	TALYS2
101	$^{11}\text{Be} + d \leftrightarrow ^{12}\text{Be} + p$	TALYS2	282	$^9\text{Li} + d \leftrightarrow ^{10}\text{Be} + n$	TALYS2
102	$^{11}\text{Be} + d \leftrightarrow ^9\text{Li} + \alpha$	TALYS2	283	$^9\text{Li} + d \leftrightarrow ^{11}\text{Be} + \gamma$	TALYS2
103	$^{11}\text{Be} + n \leftrightarrow ^{12}\text{Be} + \gamma$	Rau94 [†]	284	$^9\text{Li} + d \leftrightarrow ^8\text{Li} + t$	TALYS2
104	$^{11}\text{Be} + p \leftrightarrow ^{11}\text{B} + n$	TALYS2	285	$^9\text{Li} + p \leftrightarrow ^{10}\text{Be} + \gamma$	TALYS2
105	$^{11}\text{Be} + p \leftrightarrow ^{12}\text{B} + \gamma$	TALYS2	286	$^9\text{Li} + p \leftrightarrow ^9\text{Be} + n$	TALYS2
106	$^{11}\text{Be} + p \leftrightarrow ^{10}\text{Be} + d$	TALYS2	287	$^9\text{Li} + p \leftrightarrow ^6\text{He} + \alpha$	li7pa (borrowed)
107	$^{11}\text{Be} + p \leftrightarrow ^9\text{Be} + t$	TALYS2	288	$^9\text{Li} + p \leftrightarrow ^7\text{Li} + t$	TALYS2
108	$^{11}\text{Be} + p \leftrightarrow ^8\text{Li} + \alpha$	TALYS2	289	$^9\text{Li} + t \leftrightarrow ^{11}\text{Be} + n$	TALYS2
109	$^{11}\text{Be} + t \leftrightarrow ^{13}\text{B} + n$	TALYS2	290	$^9\text{Li} + t \leftrightarrow ^{12}\text{Be} + \gamma$	TALYS2
110	$^{11}\text{Be} + t \leftrightarrow ^{14}\text{B} + \gamma$	TALYS2	291	$^{12}\text{N} \rightarrow ^{12}\text{C} + e^+$	Aud03 [†]
111	$^{12}\text{Be} + ^3\text{He} \leftrightarrow ^{14}\text{B} + p$	TALYS2	292	$^{12}\text{N} + \alpha \leftrightarrow ^{15}\text{O} + p$	TALYS2
112	$^{12}\text{Be} + ^3\text{He} \leftrightarrow ^{11}\text{Be} + \alpha$	TALYS2	293	$^{12}\text{N} + n \leftrightarrow ^{11}\text{C} + d$	TALYS2
113	$^{12}\text{Be} + ^3\text{He} \leftrightarrow ^{14}\text{C} + n$	TALYS2	294	$^{12}\text{N} + n \leftrightarrow ^{12}\text{C} + p$	TALYS2
114	$^{12}\text{Be} + ^3\text{He} \leftrightarrow ^{15}\text{C} + \gamma$	TALYS2	295	$^{12}\text{N} + n \leftrightarrow ^{13}\text{N} + \gamma$	TALYS2
115	$^{12}\text{Be} \rightarrow ^{12}\text{B} + e^-$	Aud03 [†]	296	$^{12}\text{N} + p \leftrightarrow ^{13}\text{O} + \gamma$	TALYS2
116	$^{12}\text{Be} + \alpha \leftrightarrow ^{15}\text{C} + n$	TALYS2	297	$^{13}\text{N} \rightarrow ^{13}\text{C} + e^+$	Aud03 [†]
117	$^{12}\text{Be} + \alpha \leftrightarrow ^{16}\text{C} + \gamma$	TALYS2	298	$^{13}\text{N} + \alpha \leftrightarrow ^{17}\text{F} + \gamma$	TALYS2
118	$^{12}\text{Be} + d \leftrightarrow ^{13}\text{B} + n$	TALYS2	299	$^{13}\text{N} + n \leftrightarrow ^{12}\text{C} + d$	TALYS2
119	$^{12}\text{Be} + d \leftrightarrow ^{14}\text{B} + \gamma$	TALYS2	300	$^{13}\text{N} + n \leftrightarrow ^{13}\text{C} + p$	TALYS2
120	$^{12}\text{Be} + p \leftrightarrow ^{12}\text{B} + n$	TALYS2	301	$^{13}\text{N} + n \leftrightarrow ^{14}\text{N} + \gamma$	TALYS2
121	$^{12}\text{Be} + p \leftrightarrow ^{13}\text{B} + \gamma$	TALYS2	302	$^{13}\text{N} + p \leftrightarrow ^{14}\text{O} + \gamma$	NACRE II
122	$^{12}\text{Be} + p \leftrightarrow ^{10}\text{Be} + t$	TALYS2	303	$^{14}\text{N} + \alpha \leftrightarrow ^{18}\text{F} + \gamma$	ILCCF10
123	$^{12}\text{Be} + p \leftrightarrow ^9\text{Li} + \alpha$	TALYS2	304	$^{14}\text{N} + n \leftrightarrow ^{14}\text{C} + p$	CF88 [†]
124	$^{12}\text{Be} + t \leftrightarrow ^{14}\text{B} + n$	TALYS2	305	$^{14}\text{N} + n \leftrightarrow ^{15}\text{N} + \gamma$	TALYS2
125	$^{12}\text{Be} + t \leftrightarrow ^{15}\text{B} + \gamma$	TALYS2	306	$^{14}\text{N} + p \leftrightarrow ^{15}\text{O} + \gamma$	NACRE II
126	$^7\text{Be} + ^3\text{He} \leftrightarrow ^{10}\text{C} + \gamma$	TALYS2	307	$^{15}\text{N} + \alpha \leftrightarrow ^{19}\text{F} + \gamma$	ILCCF10
127	$^7\text{Be} + \alpha \leftrightarrow ^{11}\text{C} + \gamma$	NACRE II	308	$^{15}\text{N} + n \leftrightarrow ^{16}\text{N} + \gamma$	TALYS2
128	$^7\text{Be} + t \leftrightarrow ^{10}\text{B} + \gamma$	TALYS2	309	$^{15}\text{N} + p \leftrightarrow ^{12}\text{C} + \alpha$	NACRE II
129	$^7\text{Be} + t \leftrightarrow ^9\text{Be} + p$	CGXSV12	310	$^{15}\text{N} + p \leftrightarrow ^{16}\text{O} + \gamma$	NACRE II
130	$^9\text{Be} + ^3\text{He} \leftrightarrow ^{10}\text{B} + d$	TALYS2	311	$^{16}\text{N} \rightarrow ^{16}\text{O} + e^-$	Aud03 [†]
131	$^9\text{Be} + ^3\text{He} \leftrightarrow ^{11}\text{B} + p$	TALYS2	312	$^{17}\text{N} \rightarrow ^{16}\text{O} + n + e^-$	Aud03 [†]
132	$^9\text{Be} + ^3\text{He} \leftrightarrow ^{11}\text{C} + n$	TALYS2	313	$^{20}\text{Na} \rightarrow ^{20}\text{Ne} + e^+$	Aud03 [†]
133	$^9\text{Be} + ^3\text{He} \leftrightarrow ^{12}\text{C} + \gamma$	TALYS2	314	$^{20}\text{Na} + n \leftrightarrow ^{17}\text{F} + \alpha$	TALYS2
134	$^9\text{Be} + ^3\text{He} \leftrightarrow \alpha + \alpha + \alpha$	TALYS2	315	$^{21}\text{Na} \rightarrow ^{21}\text{Ne} + e^+$	Aud03 [†]
135	$^9\text{Be} + \alpha \leftrightarrow ^{12}\text{C} + n$	NACRE II	316	$^{22}\text{Na} \rightarrow ^{22}\text{Ne} + e^+$	Nubase [†]
136	$^9\text{Be} + \alpha \leftrightarrow ^{13}\text{C} + \gamma$	TALYS2	317	$^{18}\text{Ne} \rightarrow ^{18}\text{F} + e^+$	Aud03 [†]
137	$^9\text{Be} + d \leftrightarrow ^{10}\text{B} + n$	TALYS2	318	$^{18}\text{Ne} + n \leftrightarrow ^{18}\text{F} + p$	TALYS2
138	$^9\text{Be} + d \leftrightarrow ^{11}\text{B} + \gamma$	TALYS2	319	$^{18}\text{Ne} + n \leftrightarrow ^{15}\text{O} + \alpha$	TALYS2
139	$^9\text{Be} + d \leftrightarrow ^{10}\text{Be} + p$	TALYS2	320	$^{19}\text{Ne} \rightarrow ^{19}\text{F} + e^+$	Aud03 [†]
140	$^9\text{Be} + d \leftrightarrow ^7\text{Li} + \alpha$	TALYS2	321	$^{19}\text{Ne} + n \leftrightarrow ^{19}\text{F} + p$	TALYS2
141	$^9\text{Be} + d \leftrightarrow \alpha + \alpha + t$	TALYS2	322	$^{19}\text{Ne} + n \leftrightarrow ^{16}\text{O} + \alpha$	TALYS2
142	$^9\text{Be} + n \leftrightarrow ^{10}\text{Be} + \gamma$	Rau94 [†]	323	$^{19}\text{Ne} + p \leftrightarrow ^{20}\text{Na} + \gamma$	ILCCF10
143	$^9\text{Be} + p \leftrightarrow ^{10}\text{B} + \gamma$	NACRE II	324	$^{23}\text{Ne} \rightarrow ^{23}\text{Na} + e^-$	Aud03 [†]
144	$^9\text{Be} + p \leftrightarrow ^6\text{Li} + \alpha$	NACRE II	325	$^{13}\text{O} \rightarrow ^{13}\text{N} + e^+$	Aud03 [†]
145	$^9\text{Be} + p \leftrightarrow \alpha + \alpha + d$	NACRE II	326	$^{14}\text{O} \rightarrow ^{14}\text{N} + e^+$	Aud03 [†]
146	$^9\text{Be} + p \leftrightarrow \alpha + \alpha + p + n$	NACRE [†]	327	$^{14}\text{O} + \alpha \leftrightarrow ^{17}\text{F} + p$	Bar97C [†]
147	$^9\text{Be} + t \leftrightarrow ^{11}\text{B} + n$	TALYS2	328	$^{14}\text{O} + \alpha \leftrightarrow ^{18}\text{Ne} + \gamma$	Wie87 [†]
148	$^9\text{Be} + t \leftrightarrow ^{12}\text{B} + \gamma$	TALYS2	329	$^{14}\text{O} + n \leftrightarrow ^{11}\text{C} + \alpha$	TALYS2
149	$^9\text{Be} + t \leftrightarrow ^{10}\text{Be} + d$	TALYS2	330	$^{14}\text{O} + n \leftrightarrow ^{12}\text{C} + ^3\text{He}$	TALYS2
150	$^9\text{Be} + t \leftrightarrow ^8\text{Li} + \alpha$	TALYS2	331	$^{14}\text{O} + n \leftrightarrow ^{14}\text{N} + p$	CF88 [†]
151	$^{10}\text{C} \rightarrow ^{10}\text{B} + e^+$	Aud03 [†]	332	$^{14}\text{O} + n \leftrightarrow ^{15}\text{O} + \gamma$	TALYS2
152	$^{11}\text{C} + ^3\text{He} \leftrightarrow ^{10}\text{C} + \alpha$	TALYS2	333	$^{15}\text{O} \rightarrow ^{15}\text{N} + e^+$	Aud03 [†]
153	$^{11}\text{C} + ^3\text{He} \leftrightarrow ^{13}\text{N} + p$	TALYS2	334	$^{15}\text{O} + \alpha \leftrightarrow ^{19}\text{Ne} + \gamma$	ILCCF10
154	$^{11}\text{C} + ^3\text{He} \leftrightarrow ^{14}\text{O} + \gamma$	TALYS2	335	$^{15}\text{O} + n \leftrightarrow ^{12}\text{C} + \alpha$	TALYS2
155	$^{11}\text{C} \rightarrow ^{11}\text{B} + e^+$	Aud03 [†]	336	$^{15}\text{O} + n \leftrightarrow ^{15}\text{N} + p$	TALYS2
156	$^{11}\text{C} + \alpha \leftrightarrow ^{14}\text{N} + p$	NACRE [†]	337	$^{15}\text{O} + n \leftrightarrow ^{16}\text{O} + \gamma$	TALYS2
157	$^{11}\text{C} + \alpha \leftrightarrow ^{15}\text{O} + \gamma$	TALYS2	338	$^{16}\text{O} + \alpha \leftrightarrow ^{20}\text{Ne} + \gamma$	ILCCF10
158	$^{11}\text{C} + d \leftrightarrow ^{12}\text{C} + p$	CGXSV12	339	$^{16}\text{O} + n \leftrightarrow ^{17}\text{O} + \gamma$	Iga95 [†]
159	$^{11}\text{C} + d \leftrightarrow ^{13}\text{N} + \gamma$	TALYS2	340	$^{16}\text{O} + p \leftrightarrow ^{17}\text{F} + \gamma$	ILCCF10
160	$^{11}\text{C} + n \leftrightarrow ^{12}\text{C} + \gamma$	Rau94 [†]	341	$^{16}\text{O} + p \leftrightarrow ^{13}\text{N} + \alpha$	CF88 [†]
161	$^{11}\text{C} + n \leftrightarrow \alpha + \alpha + \alpha$	CGXSV12	342	$^{17}\text{O} + \alpha \leftrightarrow ^{20}\text{Ne} + n$	NACRE
162	$^{11}\text{C} + p \leftrightarrow ^{12}\text{N} + \gamma$	Tang03 [†]	343	$^{17}\text{O} + \alpha \leftrightarrow ^{21}\text{Ne} + \gamma$	CF88
163	$^{11}\text{C} + t \leftrightarrow ^{10}\text{B} + \alpha$	TALYS2	344	$^{17}\text{O} + n \leftrightarrow ^{14}\text{C} + \alpha$	Koe91 [†]
164	$^{11}\text{C} + t \leftrightarrow ^{11}\text{B} + ^3\text{He}$	TALYS2	345	$^{17}\text{O} + n \leftrightarrow ^{18}\text{O} + \gamma$	TALYS2
165	$^{11}\text{C} + t \leftrightarrow ^{12}\text{C} + d$	TALYS2	346	$^{17}\text{O} + p \leftrightarrow ^{18}\text{F} + \gamma$	ILCCF10
166	$^{11}\text{C} + t \leftrightarrow ^{13}\text{C} + p$	TALYS2	347	$^{17}\text{O} + p \leftrightarrow ^{14}\text{N} + \alpha$	ILCCF10
167	$^{11}\text{C} + t \leftrightarrow ^{13}\text{N} + n$	TALYS2	348	$^{18}\text{O} + \alpha \leftrightarrow ^{22}\text{Ne} + \gamma$	ILCCF10
168	$^{11}\text{C} + t \leftrightarrow ^{14}\text{N} + \gamma$	TALYS2	349	$^{18}\text{O} + n \leftrightarrow ^{19}\text{O} + \gamma$	TALYS2

table 3 continued

#	Reaction	Source	#	Reaction	Source
169	$^{12}\text{C} + ^3\text{He} \leftrightarrow ^{11}\text{C} + \alpha$	TALYS2	350	$^{18}\text{O} + p \leftrightarrow ^{19}\text{F} + \gamma$	ILCCF10
170	$^{12}\text{C} + ^3\text{He} \leftrightarrow ^{14}\text{N} + p$	TALYS2	351	$^{18}\text{O} + p \leftrightarrow ^{15}\text{N} + \alpha$	ILCCF10
171	$^{12}\text{C} + ^3\text{He} \leftrightarrow ^{15}\text{O} + \gamma$	TALYS2	352	$^{19}\text{O} \rightarrow ^{19}\text{F} + e^-$	Aud03 [†]
172	$^{12}\text{C} + \alpha \leftrightarrow ^{16}\text{O} + \gamma$	NACRE II	353	$^{19}\text{O} + \alpha \leftrightarrow ^{22}\text{Ne} + n$	TALYS2
173	$^{12}\text{C} + d \leftrightarrow ^{13}\text{C} + p$	TALYS2	354	$^{19}\text{O} + \alpha \leftrightarrow ^{23}\text{Ne} + \gamma$	TALYS2
174	$^{12}\text{C} + d \leftrightarrow ^{14}\text{N} + \gamma$	TALYS2	355	$^{19}\text{O} + n \leftrightarrow ^{20}\text{O} + \gamma$	TALYS2
175	$^{12}\text{C} + n \leftrightarrow ^{13}\text{C} + \gamma$	TALYS2	356	$^{19}\text{O} + p \leftrightarrow ^{19}\text{F} + n$	TALYS2
176	$^{12}\text{C} + p \leftrightarrow ^{13}\text{N} + \gamma$	NACRE II	357	$^{19}\text{O} + p \leftrightarrow ^{20}\text{F} + \gamma$	TALYS2
177	$^{12}\text{C} + t \leftrightarrow ^{11}\text{B} + \alpha$	TALYS2	358	$^{19}\text{O} + p \leftrightarrow ^{16}\text{N} + \alpha$	TALYS2
178	$^{12}\text{C} + t \leftrightarrow ^{14}\text{C} + p$	TALYS2	359	$^{20}\text{O} \rightarrow ^{20}\text{F} + e^-$	Aud03 [†]
179	$^{12}\text{C} + t \leftrightarrow ^{14}\text{N} + n$	TALYS2	360	$\alpha + \alpha + \alpha \leftrightarrow ^{12}\text{C} + \gamma + \gamma$	NACRE [†]
180	$^{12}\text{C} + t \leftrightarrow ^{15}\text{N} + \gamma$	TALYS2	361	$\alpha + \alpha + n \leftrightarrow ^9\text{Be} + \gamma$	NACRE [†]
181	$^{13}\text{C} + ^3\text{He} \leftrightarrow ^{12}\text{C} + \alpha$	TALYS2			

F Nuclide data

The nuclides tracked by the networks and their data are loaded from `nuclides.csv` (under `data/csv/`): the neutron and proton numbers (N, Z), the mass number $A = N + Z$, the spin J (degeneracy $g_i = 2J + 1$), and the atomic mass excess Δ . The latter give the binding energies $B_i = Z_i m_p + (A_i - Z_i) m_n - m_i$ and the reaction Q -values that fix the NSE abundances and the detailed-balance coefficients of App. D. Table 4 lists all 59 tracked nuclides.

Table 4: Nuclide data (`nuclides.csv`): mass number $A = N + Z$, charge Z , neutron number N , atomic mass excess Δ [MeV] from AME/NUBASE [11], binding energy $B = Z \Delta_{\text{H}} + N \Delta_{\text{n}} - \Delta$ [MeV], and spin J .

Nuc.	A	Z	N	Δ	B	J	Nuc.	A	Z	N	Δ	B	J
n	1	0	1	8.071	0.000	0.5	^{16}C	16	6	10	13.694	110.753	0
p	1	1	0	7.289	0.000	0.5	^{12}N	12	7	5	17.338	74.041	1
^2H	2	1	1	13.136	2.225	1	^{13}N	13	7	6	5.345	94.105	0.5
^3H	3	1	2	14.950	8.482	0.5	^{14}N	14	7	7	2.863	104.659	1
^3He	3	2	1	14.931	7.718	0.5	^{15}N	15	7	8	0.101	115.492	0.5
^4He	4	2	2	2.425	28.296	0	^{16}N	16	7	9	5.684	117.981	2
^6He	6	2	4	17.592	29.271	0	^{17}N	17	7	10	7.870	123.866	0.5
^6Li	6	3	3	14.087	31.994	1	^{13}O	13	8	5	23.115	75.553	1.5
^7Li	7	3	4	14.907	39.245	1.5	^{14}O	14	8	6	8.008	98.732	0
^8Li	8	3	5	20.946	41.278	2	^{15}O	15	8	7	2.856	111.955	0.5
^9Li	9	3	6	24.955	45.340	1.5	^{16}O	16	8	8	-4.737	127.619	0
^7Be	7	4	3	15.769	37.601	1.5	^{17}O	17	8	9	-0.809	131.762	2.5
^9Be	9	4	5	11.348	58.164	1.5	^{18}O	18	8	10	-0.783	139.808	0
^{10}Be	10	4	6	12.607	64.976	0	^{19}O	19	8	11	3.333	143.763	2.5
^{11}Be	11	4	7	20.177	65.478	0.5	^{20}O	20	8	12	3.796	151.371	0
^{12}Be	12	4	8	25.078	68.649	0	^{17}F	17	9	8	1.952	128.220	2.5
^8B	8	5	3	22.922	37.737	2	^{18}F	18	9	9	0.873	137.370	1
^{10}B	10	5	5	12.051	64.751	3	^{19}F	19	9	10	-1.487	147.801	0.5
^{11}B	11	5	6	8.668	76.205	1.5	^{20}F	20	9	11	-0.017	154.403	2
^{12}B	12	5	7	13.369	79.575	1	^{18}Ne	18	10	8	5.318	132.143	0
^{13}B	13	5	8	16.562	84.454	1.5	^{19}Ne	19	10	9	1.752	143.780	0.5
^{14}B	14	5	9	23.664	85.423	2	^{20}Ne	20	10	10	-7.042	160.645	0
^{15}B	15	5	10	28.957	88.201	1.5	^{21}Ne	21	10	11	-5.732	167.406	1.5
^9C	9	6	3	28.911	39.037	1.5	^{22}Ne	22	10	12	-8.025	177.770	0
^{10}C	10	6	4	15.699	60.320	0	^{23}Ne	23	10	13	-5.154	182.971	2.5
^{11}C	11	6	5	10.649	73.441	1.5	^{20}Na	20	11	9	6.851	145.970	2
^{12}C	12	6	6	0.000	92.162	0	^{21}Na	21	11	10	-2.185	163.077	1.5
^{13}C	13	6	7	3.125	97.108	0.5	^{22}Na	22	11	11	-5.181	174.145	3
^{14}C	14	6	8	3.020	105.284	0	^{23}Na	23	11	12	-9.530	186.564	1.5

table 4 continued

Nuc.	<i>A</i>	<i>Z</i>	<i>N</i>	Δ	<i>B</i>	<i>J</i>	Nuc.	<i>A</i>	<i>Z</i>	<i>N</i>	Δ	<i>B</i>	<i>J</i>
¹⁵ C	15	6	9	9.873	106.503	0.5							

G Reproducing the figures

Every figure in this document is generated by the notebook `manual/primat_doc_figures.ipynb`, which writes PDFs into `manual/figures/`. To refresh them after changing the code, run the notebook top to bottom, or non-interactively:

```
jupyter nbconvert --to notebook --execute --inplace primat_doc_figures.ipynb
```

Similarly, App. E’s reaction-list table is regenerated (from the current network/rate-table data, not from a run) by

```
python manual/generate_tab_reactions.py
```

Then recompile the document. The notebook uses only the public primat API (the Python back-end, so that the intermediate background/weak-rate/reaction-rate attributes of §1 are available) and doubles as worked examples for extracting the background history, weak rates, reaction rates and abundance trajectories from a run.

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